AI-Assisted Criminal Investigations: Enhancing Testimony Analysis and Case Correlation

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Final Report

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Declaration

I declare that this is my own work, and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute my dissertation in whole or part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as article or books)

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Date: 29/8/2025

The above candidates has carried out research for the bachelor’s degree Dissertation under my supervision.

Signature of the supervisor: Date:

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Abstract

This research addresses critical limitations in traditional criminal investigations by developing an AI-assisted system that enhances testimony analysis and dynamic interrogation processes. Conventional interrogation methods rely extensively on investigator expertise, which can be affected by cognitive biases, fatigue, and subjective judgment, leading to inefficiencies and compromised accuracy. Leveraging advances in Natural Language Processing (NLP) and transformer-based Large Language Models (LLMs), this system introduces dynamic question generation tailored in real-time to suspect responses. It integrates real-time analysis of testimony to detect inconsistencies, contradictions, and missing information, thereby providing investigators with actionable and contextually relevant follow-up questions. The system further incorporates multimodal behavioral analytics including emotion and stress indicators from speech and facial cues to enrich interrogation insights. Architected as a scalable full-stack application with React/Next.js frontend, MongoDB backend, and APIs interfacing with LLMs, the platform supports secure, multi-session interrogations with an intuitive investigator interface. Comprehensive feasibility studies, field visits, and data collection from Sri Lankan law enforcement ensured alignment with operational, legal, and ethical requirements. Experimental evaluation demonstrated the LLM's superior semantic relevance for question generation against expert-curated benchmarks. The system maintains ethical safeguards such as human oversight, data anonymization, and bias monitoring to ensure fairness and judicial compliance. This AI-assisted interrogation assistant promises to enhance investigation efficiency, accuracy, and ethical standards while accelerating case resolution and strengthening the criminal justice process

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Abbreviations

AI – Artificial Intelligence

API – Application Programming Interface

CNN – Convolutional Neural Network

GCP – Google Cloud Platform

GDPR – General Data Protection Regulation

GPU – Graphics Processing Unit

IDE – Integrated Development Environment

JSON – JavaScript Object Notation

LLM – Large Language Model

MFA – Multi-Factor Authentication

NFR – Non-Functional Requirement

NLP – Natural Language Processing

OCR – Optical Character Recognition

PII – Personally Identifiable Information

RBAC – Role-Based Access Control

RNN – Recurrent Neural Network

SaaS – Software as a Service

SBERT – Sentence-BERT (Sentence Bidirectional Encoder Representations from Transformers)

SDK – Software Development Kit

SQL – Structured Query Language

UI – User Interface

UX – User Experience

WCAG – Web Content Accessibility Guidelines

1. Introduction

1.1. Background

Traditional criminal investigations rely heavily on manual processes, where investigators record incident details in physical, often handwritten documents. These investigations are emotionally charged and demand precise and consistent documentation. Tasks such as interviewing victims and witnesses, interrogating suspects, interpreting behavioral cues, and collecting evidence to build a case are typically left to the discretion of individual investigators. Moreover, investigators must sift through large volumes of unstructured data, cross-reference multiple testimonies, and identify connections between new and historical cases. Such practices expose several drawbacks, including inconsistent case recording, misplaced or incomplete records, and delays in establishing meaningful links. The process is also shaped by subjective factors such as the investigator's training, experience, judgment, and rank. Consequently, critical behavioral cues may be missed, and important insights overlooked, particularly when correlating evidence, testimonies, and past case data is done manually.

Recent advancements in Artificial Intelligence (AI) have introduced promising solutions to these limitations. Natural Language Processing (NLP) and Large Language Models (LLMs) can automatically extract and structure information from unstructured documents, cross-reference case details, generate contextually relevant questions, and analyze responses during interrogations. Machine Learning (ML) techniques can correlate past data with current cases, uncovering hidden patterns and critical insights. Additionally, Computer Vision combined with ML enables the recognition of emotional cues in suspects' facial expressions, while Internet of Things (IoT) devices can capture physiological indicators such as heart rate, providing further evidence of stress or deception.

The proposed research addresses these drawbacks by leveraging AI techniques within an integrated platform that assists investigators throughout the investigation process. The system is designed to (1) automate data extraction from police reports and testimonies using LLMs, (2) detect stress and deception through facial expression and heart rate analysis, (3) analyze responses and dynamically generate follow-up questions during interrogations, and (4) identify patterns and correlations with historical case data. By fusing behavioral analysis with advanced data mining into a unified investigative tool, this research aims to enhance the efficiency, accuracy, and objectivity of criminal investigations, thereby strengthening the capabilities of modern criminal justice systems.

1.2.Literature Review

The adoption of Artificial Intelligence (AI) in criminal investigations has been driven by the increasing need to process large volumes of unstructured testimonies, police reports, and case records with greater speed, accuracy, and consistency. Traditional manual workflows are prone to human error, investigator fatigue, and subjective bias, which often lead to inefficiencies and overlooked evidence [1], [2].

Earlier technological integrations in law enforcement focused heavily on Optical Character Recognition (OCR) to digitize physical police reports, followed by Natural Language Processing (NLP) for entity recognition and classification. However, OCR-based systems have demonstrated limitations when processing handwritten, noisy, or multilingual documents [3]. Greif et al. [4] and Patel [5] highlight that multimodal large language models (mLLMs) outperform traditional OCR by jointly leveraging textual and visual context, enabling direct entity extraction from scanned images and handwritten pages. Generative AI-based OCR has also introduced contextual reasoning, few-shot adaptability, and robust multilingual processing, particularly relevant for underrepresented languages such as Sinhala and Tamil [6]. Despite these advances, end-to-end legal document extraction in these languages remains underexplored.

Testimonies and incident narratives represent a rich but underutilized source of evidence in criminal investigations. Early computational approaches relied on keyword searches and lexical overlap to link similar cases. Johnson et al. [7] and Lee et al. [8] demonstrated that clustering and neural networks could reveal hidden trends across offender profiles and geographic distributions. With the advent of transformer-based models such as BERT, testimony analysis has shifted toward semantic similarity, enabling the detection of contradictions and narrative overlaps beyond keyword matching [9]. Smith et al. [10] confirmed that semantic embeddings outperform traditional search techniques by uncovering hidden relationships in police testimonies. However, most prior studies remain offline prototypes without real-time operational deployment.

Human behaviour and stress responses have long been used as indicators of deception and testimony credibility. Conventional tools such as polygraphs and standalone voice stress analyzers have been criticized for their unreliability and vulnerability to countermeasures [11]. Recent research on **facial** emotion recognition (FER) demonstrates that CNN-based models can detect subtle emotional cues, with datasets such as FER-2013 and AffectNet enabling recognition of micro-expressions [12], [13]. Parallel work on speech emotion recognition (SER) leverages hybrid CNN–RNN and attention-based models to identify vocal stress patterns in forensic interviews [14], [15]. Physiological monitoring, particularly heart rate variability (HRV) analysis from ECG signals, provides an additional biological marker of stress and deception [16], [17]. While these unimodal methods show promise individually, Das et al. [18] and Baltrūnienė [19] emphasize that multimodal fusion combining FER, SER, and HRV yields more robust and context-aware behavioural profiles. Nonetheless, real-time multimodal interrogation support systems remain scarce in literature.

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Most AI-driven investigative systems have historically emphasized offender profiling, often neglecting victim vulnerability analysis. Criminological studies highlight the importance of demographic and situational factors in predicting risks of repeat victimization or escalation [24]. Campbell [25] and Baltrūnienė [19] underscore the lack of structured victim risk modules in current AI systems. Recent frameworks attempt to address this by incorporating heuristic scoring for factors such as age, injury severity, and recurrence of violence [10]. Alongside risk scoring, explainability remains a crucial factor in ensuring adoption in law enforcement. Black-box models are often distrusted by investigators and courts; therefore, interpretable outputs and audit trails are essential for operational deployment [26].

1.3. Research Gap

Traditional Optical Character Recognition (OCR)-based frameworks remain limited in handling handwritten reports, noisy scans, and multilingual content. While multimodal Large Language Models (LLMs) offer OCR-free processing and contextual extraction, their application in criminal justice—particularly for Sinhala and Tamil legal documents—is minimal. Existing systems also fail to unify document extraction with real-time synchronization into investigator-friendly platforms.

Current deception detection and testimony evaluation tools are predominantly unimodal, focusing on either facial emotion recognition, speech emotion recognition, or physiological monitoring in isolation. Research shows that combining modalities increases robustness, yet real-time multimodal interrogation systems with synchronized FER, SER, and HRV are rarely explored. Furthermore, most prototypes remain laboratory-bound, lacking investigator-facing dashboards or operational usability

Existing AI systems often emphasize offender profiling, with limited attention to victim vulnerability assessment. While clustering and NLP methods have been used for case linking, few platforms integrate structured demographic attributes with semantic testimony similarity into a unified pipeline. Explainability is another major gap: investigators and courts remain reluctant to trust black-box models without transparent reasoning or contextual evidence.

Interrogation support technologies remain narrow in scope, focusing on static question banks, polygraph testing, or isolated deception detection. There is a lack of adaptive AI-driven question generation that dynamically responds to suspect testimonies. Moreover, integration of behavioral cues (stress, emotion, inconsistencies) with interrogation flows is largely unexplored. Current research prototypes rarely address deployment challenges such as investigator usability, multilingual support, and ethical/legal compliance.

Features	**Research Paper [1]**	**Research Paper [2]**	**Research Paper [3]**	**Research Paper [4]**	**Victim Risk (A)**	**Dynamic Q&A (B)**	**Data Collection (C)**	**Behavioural Analysis (D)**	**Proposed Integrated System**
Integration of Structured + Unstructured Data	✗	✗	✓	✗	✗	✗	✓	✗	✓
Real-Time Testimony Similarity (LLM/BERT)	✗	✓	✗	✗	✗	✓	✗	✗	✓
Automated Incident	✗	✓	✓	✗	✓	✗	✓	✗	✓

Correlation									
Victim-Centered Risk Assessment	✗	✗	✗	✗	✓	✗	✗	✗	✓
Explainability of Predictions	✗	✓	✗	✗	✗	✓	✗	✗	✓
Multilingual & Handwritten Document Support	✗	✗	✓	✗	✗	✗	✓	✗	✓
Behavioural & Physiological Stress Detection	✗	✗	✗	✓	✗	✗	✗	✓	✓
Integrated Investigator Dashboard	✗	✗	✓	✓	✗	✓	✓	✓	✓

Table 1 : Research Gap

1.4.Research Problem

The research problem addresses the significant limitations and inefficiencies in traditional criminal investigation processes, which often rely heavily on manual efforts and subjective human interpretation. Investigators are responsible for collecting, organizing, and analyzing large volumes of case data, such as testimonies, incident reports, and evidence. This manual process is prone to human error, including misinterpretations, biases, inconsistencies in judgment, and fatigue, all of which can lead to inaccuracies, overlooked details, and delayed case resolution

A major difficulty lies in the fragmented nature of existing investigative methods. The process of cross-referencing testimonies, correlating new incidents with historical cases, and identifying hidden behavioral patterns is typically slow and manual, which prevents timely identification of trends or contradictions. Traditional OCR and NLP systems provide some automation but are often limited in accuracy, especially with multilingual or handwritten reports, and lack deeper semantic reasoning. Similarly, testimony credibility assessments still depend largely on investigator intuition or unimodal tools (facial, vocal, or physiological analysis in isolation), which are vulnerable to noise, bias, and fatigue-related errors.

Interrogations also remain constrained by static questioning frameworks and investigator expertise. Without dynamic support, questioning strategies often fail to adapt to evolving suspect responses. This not only risks missing contradictions or critical insights but also amplifies cognitive bias and reduces efficiency. Furthermore, victim-centered considerations such as vulnerability assessment remain underrepresented in current investigative platforms, while explainability of AI outputs is rarely prioritized, undermining trust in technology-assisted judgments

Currently, very few systems are available globally to assist in such investigations, and the few that exist are narrow in scope. Most focus on isolated tasks, such as lie detection, biometric analysis, or basic data management, and lack the integration of advanced multimodal analytics and dynamic question generation into a comprehensive decision-support framework. These gaps prevent investigators from fully leveraging both structured and unstructured evidence in real time.

This combination of human error, inefficiency, and the lack of advanced, integrated systems underscores the need for a more robust, AI-driven solution to enhance the accuracy, efficiency, and effectiveness of criminal investigations. By integrating automated multilingual data extraction, multimodal behavioral analysis, semantic testimony similarity, case correlation, victim risk assessment, and dynamic question generation, the proposed system aims to modernize investigative workflows, provide real-time support to investigators, and strengthen the fairness and reliability of criminal justice processes.

1.5.Research Objectives

1.5.1. Main Objective

Develop an AI-driven system that enhances the efficiency and accuracy of criminal investigations. By automating data collection, analyzing testimonies in real-time, and identifying patterns across incidents, the system reduces human error and provides investigators with actionable insights. The integration of video, heart rate, audio, and witness data, along with machine learning, aims to streamline investigative processes and improve decision-making, accelerating case resolution.

1.5.2. Specific Objectives

- Automate the extraction, organization, and management of case data using AI technologies.
    - Automate the extraction of structured and unstructured data from police reports, testimonies, and incident files using LLMs and NLP pipelines to reduce reliance on traditional OCR systems.
    - Ensure support for multilingual document processing (Sinhala, Tamil, English) to address regional diversity and improve accessibility of testimony analysis.
    - Implement secure and scalable data storage solutions (MongoDB, cloud integration) for centralized case management with audit trails and controlled access.
    - Enable real-time synchronization and preprocessing of digitized reports, ensuring investigators have consistent and up-to-date case records for decision-making.
- Develop a multi-modal system for real-time analysis of emotional and physiological behavior during suspect interrogations.
    - Design and implement a Facial Emotion Recognition (FER) system using CNN-based deep learning models trained on benchmark datasets (e.g., FER-2013).
    - Develop a Speech Emotion Recognition (SER**)** module leveraging hybrid CNN–BiLSTM models with attention mechanisms, trained on datasets such as RAVDESS, to capture stress and emotional variations in vocal signals.
    - Integrate Heart Rate Variability (HRV) analysis using ECG signals (AD8232 sensor) and physiological stress indicators to complement behavioral analysis.
    - Build **a** fusion framework that synchronizes FER, SER, and HRV signals to produce a unified Stress Index, increasing the robustness of deception and stress detection.
    - Provide an investigator-facing dashboard that visualizes multimodal outputs in real time, offering clear, interpretable behavioral insights without overwhelming users.
- Implement a dynamic, real-time question generation and response analysis system to enhance interrogation efficiency.

- Develop an LLM-powered module (Gemini, GPT-4) capable of dynamically generating contextually relevant follow-up questions during interrogations.
- Implement a real-time response analysis engine to detect inconsistencies, contradictions, and missing information in suspect testimonies.
- Design an interactive investigator interface that allows review, approval, or modification of AI-generated questions to maintain human oversight.
- Establish a feedback-driven reinforcement mechanism, enabling investigators' evaluations to refine model outputs over time for improved adaptability.
- Ensure integration of multimodal behavioral cues (stress, emotional fluctuations) into the questioning flow, enriching interrogation strategies with contextual insights.

- Leverage machine learning algorithms to identify patterns, correlations, and discrepancies between new incidents and historical case data.
  - Design and train a Random Forest classifier on structured case data (e.g., victim and perpetrator demographics, type of violence, evidence indicators) to predict criminal responsibility.
  - Implement semantic similarity analysis of testimonies using transformer-based models (e.g., SBERT/BERT embeddings) to detect hidden overlaps, contradictions, or repeated patterns in narratives.
  - Build an incident correlation engine combining semantic embeddings with lightweight keyword matching to identify both deep narrative connections and surface-level similarities.
  - Develop a victim risk assessment module using heuristic scoring (age, gender, recurrence, injuries, relationship to perpetrator) to highlight levels of vulnerability.
  - Provide transparent, explainable outputs with clear reasoning, visualized verdicts, and links to historical cases through an investigator-friendly dashboard.

2. Methodology

2.1. Requirements Gathering

The requirements gathering process formed the foundation for the design and development of the AI-Assisted Criminal Investigations: Enhancing Testimony Analysis and Case Correlation System. To ensure that the system addressed real investigative challenges while maintaining technical feasibility, a multi-pronged approach was adopted, combining literature review, field visits, structured interviews, dataset analysis, and consultations with domain experts.

2.1.1. Conducting Literature Survey

An extensive review of prior research was conducted across multiple domains relevant to the project, including natural language processing, multimodal behavioural analysis, case correlation, risk assessment, and dynamic question generation. The survey revealed that while deep learning models for facial emotion recognition (FER), speech emotion recognition (SER), and heart rate variability (HRV) analysis had shown promise individually, few systems successfully integrated these modalities into an operational interrogation support framework. Similarly, testimony analysis using transformer-based NLP models such as BERT and SBERT demonstrated potential for semantic similarity detection, but most studies lacked real-time, investigator-facing applications. Existing platforms also paid limited attention to victim vulnerability assessment or to the practical usability of AI systems in law enforcement contexts. These findings highlighted the need for a unified, modular system that bridges academic advances and operational deployment.

2.1.2. Field Visit

To ensure the proposed **AI-Assisted Criminal Investigations System** addressed real-world investigative needs, multiple field visits were conducted to police stations and law enforcement offices in Sri Lanka. These visits, complemented by structured interviews with serving officers and external supervisors with decades of experience in policing, provided critical insights into existing workflows, limitations, and expectations of technological support.

1. Case Documentation and Data Management

   Investigators highlighted the continued reliance on manual record-keeping **systems**, including handwritten police reports and testimony notes. The process of organizing, retrieving, and cross-referencing case files was described as time-consuming and error-prone, especially when dealing with multilingual records in Sinhala, Tamil, and English. Officers confirmed that important details were often missed due to inconsistent reporting formats and the absence of automated digitization and extraction tools. These observations validated the need for OCR-free, multilingual AI-driven data extraction and structured case management solutions.

2. Testimony Evaluation and Behavioural Observation

   During interrogation sessions observed in the field, investigators relied primarily on personal expertise to interpret behavioural cues, such as facial expressions, tone of voice, and body language. Officers acknowledged that this subjective approach was vulnerable to bias, fatigue, and inconsistency, particularly during long or complex investigations. Crucially, visits revealed the absence of any technological tools to support real-time monitoring of emotional or physiological signals (e.g., stress detection through facial, vocal, or heart rate indicators). This reinforced the importance of integrating multimodal behavioural and physiological analysis into the system.

3. Cross-Case Correlation and Victim Risk Assessment

   Field interactions revealed significant challenges in linking new cases with historical records. Investigators explained that testimonies and case details were often manually compared, making it difficult to detect contradictions, repeated offender behaviour, or hidden connections between incidents. Furthermore, little systematic support was available for assessing victim vulnerability, even in high-risk domestic violence or repeat abuse cases. Officers noted that such assessments were carried out informally, relying on individual judgment rather than structured criteria. These insights highlighted the need for semantic testimony similarity analysis, automated case correlation, and a victim risk assessment module to improve both investigative accuracy and victim protection.

4. Interrogation and Questioning Practices

   The field visits also underscored the limitations of current interrogation practices. Investigators typically used static sets of questions and depended heavily on personal intuition to pursue contradictions or missing details. They reported difficulties in adapting questioning strategies in real time as suspect responses evolved, leading to inefficiencies and missed opportunities to uncover critical evidence. Manual note-taking and post-interview analysis further increased the likelihood of delays and oversight. Officers expressed interest in tools that could provide dynamic, AI-generated follow-up questions and automated documentation of interrogation sessions.

2.1.3. Data Collection

The development of the AI-Assisted Criminal Investigations System required a comprehensive and multi-source data collection strategy to support its four core modules: automated data extraction, multimodal behavioural analysis, case correlation and risk assessment, and dynamic question generation. Data was gathered from both real-world police records and benchmark datasets, ensuring that the system could generalize across controlled academic corpora and operational investigative contexts.

1. Case Reports and Testimonies

Anonymized police reports and case records were collected through field visits to law enforcement agencies. These included handwritten statements, multilingual reports (Sinhala, Tamil, English), and incident summaries, which served as the foundation for developing the document extraction and testimony analysis modules. To ensure privacy and compliance, personally identifiable information (PII) was removed, and synthetic case reports were generated to simulate sensitive scenarios. This dataset was used for:

- Training OCR-free NLP pipelines for automated entity extraction.
- Building LLM prompts for testimony summarization and contradiction detection.
- Supporting the dynamic question generation module, which required real-world testimony contexts to test adaptive questioning.

2. Behavioural and Physiological Datasets

For the Multimodal Behavioural and Physiological Analysis (MBPA) component, multiple benchmark datasets were adopted:

- Facial Emotion Recognition (FER): FER-2013 dataset (~35,000 labeled facial images across 7 emotions).
- Speech Emotion Recognition (SER): RAVDESS dataset (1,440 audio samples covering 8 emotions with varying intensity).
- Physiological Monitoring (HRV/ECG): A hybrid approach combining (i) experimental ECG recordings collected using the AD8232 heart rate sensor in controlled conditions, and (ii) publicly available datasets such as YAAD, which provide labeled ECG/GSR recordings of emotional states.

To improve generalization, preprocessing and augmentation techniques were applied, including noise injection, pitch/time modifications (SER), facial alignment and grayscale normalization (FER), and artifact removal with band-pass filtering (ECG). These steps ensured robustness under interrogation-room conditions, where data quality can be inconsistent.

3. Structured Case Data for Correlation and Risk Assessment

A structured dataset containing 500 labeled criminal cases was prepared for the case correlation and victim risk assessment module. Each record included:

- Victim and perpetrator demographics (age, gender).
- Case attributes (type of violence, recurrence, evidence presence).
- Relationships between victims and perpetrators.
- Case outcomes (labels for classification).

Derived features such as age difference and evidence indicators were engineered to enhance model performance. This dataset supported training of a Random Forest classifier for offender likelihood prediction and heuristic scoring for victim vulnerability assessment.

4. Interrogation Data for Dynamic Questioning

To support the dynamic question generation and response analysis module, a dataset of anonymized and synthetic interrogation transcripts was curated. Collected police reports and testimonies were reformatted into question–answer pairs, allowing Large Language Models (Gemini, GPT-4) to be evaluated for follow-up question generation and semantic inconsistency detection. Synthetic datasets were also created to test rare or edge-case scenarios, ensuring broader coverage. This dataset was further validated against expert-curated question banks to benchmark semantic accuracy.

2.2.Feasibility Study

The feasibility of the proposed **AI-Assisted Criminal Investigations System** was carefully examined across four dimensions: **technical, operational, legal, and ethical**. This analysis ensured that the system design not only leverages state-of-the-art technology but also aligns with the practical, regulatory, and societal requirements of law enforcement.

1. Technical Feasibility

The system was found to be technically feasible due to the maturity of available AI models, datasets, and software infrastructures:

- **Automated Data Extraction and Management**: Transformer-based Large Language Models (LLMs) such as **Gemini 2.0** provide OCR-free document understanding, enabling accurate extraction from **handwritten, multilingual police reports**. MongoDB and Google Cloud APIs ensure secure, scalable storage and access.
- **Multimodal Behavioural Analysis (MBPA)**: CNN-based **Facial Emotion Recognition (FER)** models, CNN–BiLSTM with attention for **Speech Emotion Recognition (SER)**, and **ECG-based HRV analysis** have all been validated on benchmark datasets (FER-2013, RAVDESS, YAAD). Containerized **FastAPI microservices** allow independent deployment of each unimodal model, ensuring modularity and real-time performance.
- **Case Correlation and Risk Assessment**: Machine learning classifiers such as **Random Forests (Scikit-learn)** and **semantic embeddings (SBERT/BERT)** provide robust predictive and similarity analysis. Cloud-hosted **MongoDB Atlas** supports storage of structured/unstructured case data, while FastAPI ensures smooth backend orchestration.
- **Dynamic Question Generation and Response Analysis**: Transformer-based LLMs (Gemini, GPT-4) have demonstrated strong capabilities in contextual question generation and semantic inconsistency detection. The **Next.js + TypeScript frontend** and scalable microservices architecture support real-time interrogation dashboards.

Together, these technologies confirm the technical feasibility of implementing a real-time, scalable, and integrated investigative decision-support system.

2. Operational Feasibility

Operational feasibility was validated through field visits, interviews, and feedback from law enforcement personnel:

- Investigators confirmed that current workflows are hindered by manual documentation, fragmented case correlation, and subjective testimony analysis.
- Officers expressed the need for investigator-friendly dashboards that present clear, actionable outputs without requiring advanced technical expertise.
- The modular design ensures seamless integration into existing workflows without disrupting established practices: officers continue to log cases as usual, while the system automatically processes and analyzes data in the background.

- The multimodal interrogation assistant (MBPA + Dynamic Questioning) was considered particularly valuable for reducing fatigue, providing objective behavioural insights, and improving efficiency of interviews.

These validations confirm that the system is operationally feasible, addressing practical challenges while remaining aligned with investigative norms.

3. Legal Feasibility

Given the sensitivity of interrogation and case data, strict legal compliance is essential. The system incorporates several measures:

- Data Protection and Privacy: All collected case reports and biometric data are anonymized, encrypted, and stored securely, ensuring compliance with Sri Lankan law and international data protection standards (e.g., GDPR).
- Auditability: Every AI-generated output is logged with timestamps, model versioning, and contributing factors, creating transparent audit trails for judicial review.
- Explainability: Outputs are presented in interpretable formats (e.g., similarity scores, victim risk levels, multimodal stress indices) that can withstand scrutiny in court.

These measures ensure the system's legal admissibility and acceptance within law enforcement and judicial contexts.

4. Ethical Feasibility

The ethical feasibility of the system was a central concern, given its deployment in sensitive interrogation and investigative contexts:

- Human-in-the-Loop Oversight: The system is designed as a decision-support tool, not a replacement for investigator judgment. AI-generated questions, risk assessments, and predictions are provided as recommendations, with investigators retaining full authority.
- Bias Mitigation: Training datasets were balanced where possible, and augmentation techniques were applied to minimize demographic bias in FER, SER, and testimony analysis. Continuous monitoring of model performance ensures fairness across gender, age, and ethnicity.

- Non-Coercive Use: Dynamic question generation and stress analysis modules are designed to highlight inconsistencies and emotional cues without pressuring suspects or victims, ensuring ethical interrogation practices.
- Transparency and Trust: Outputs include modality contributions (e.g., “65% HRV, 35% SER/FER”) and explanations of predictions, fostering investigator trust and reducing the “black box” problem.

2.3.System Design

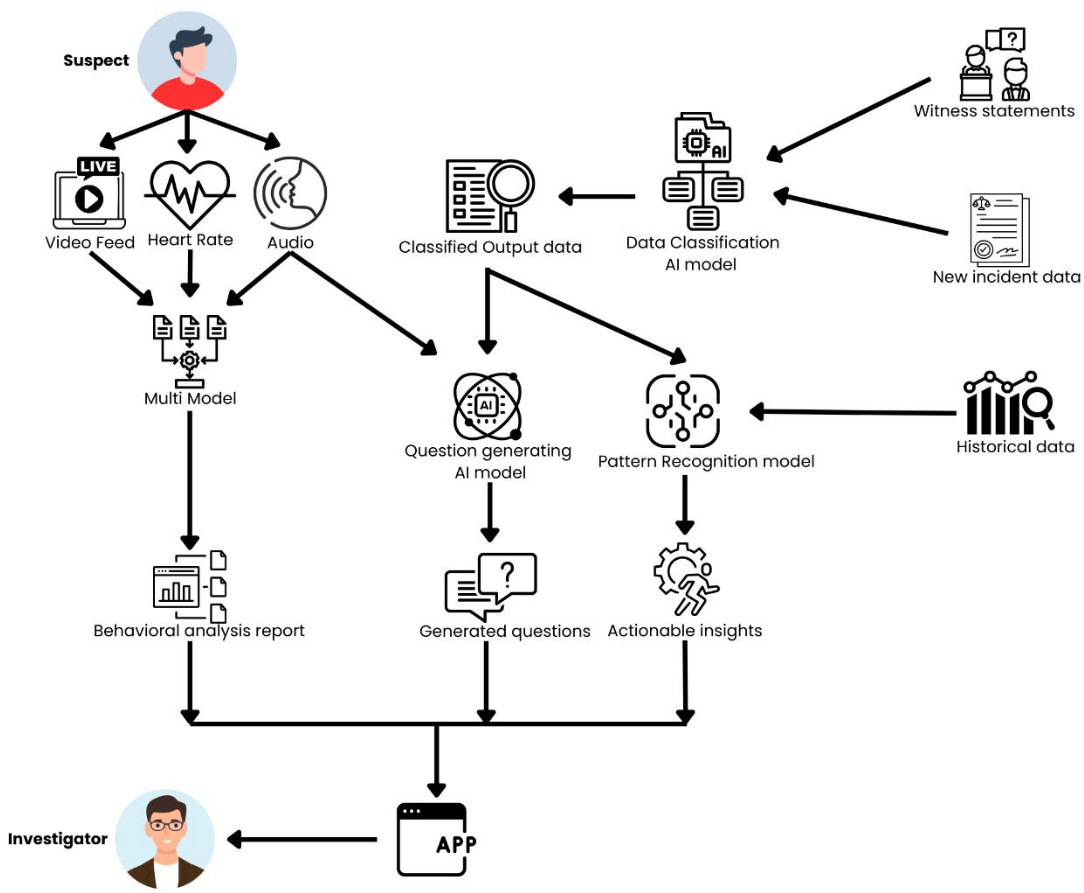


Figure 1: System Design

The AI-Assisted Criminal Investigations System is designed as a modular, microservices-based platform that integrates four key components: automated data collection and management, multimodal behavioural and physiological analysis, case correlation and risk assessment, and dynamic question generation with response analysis. Together, these modules form a unified decision-support system aimed at enhancing the efficiency, accuracy, and fairness of criminal investigations.

2.3.1. Automated Data Collection and Management

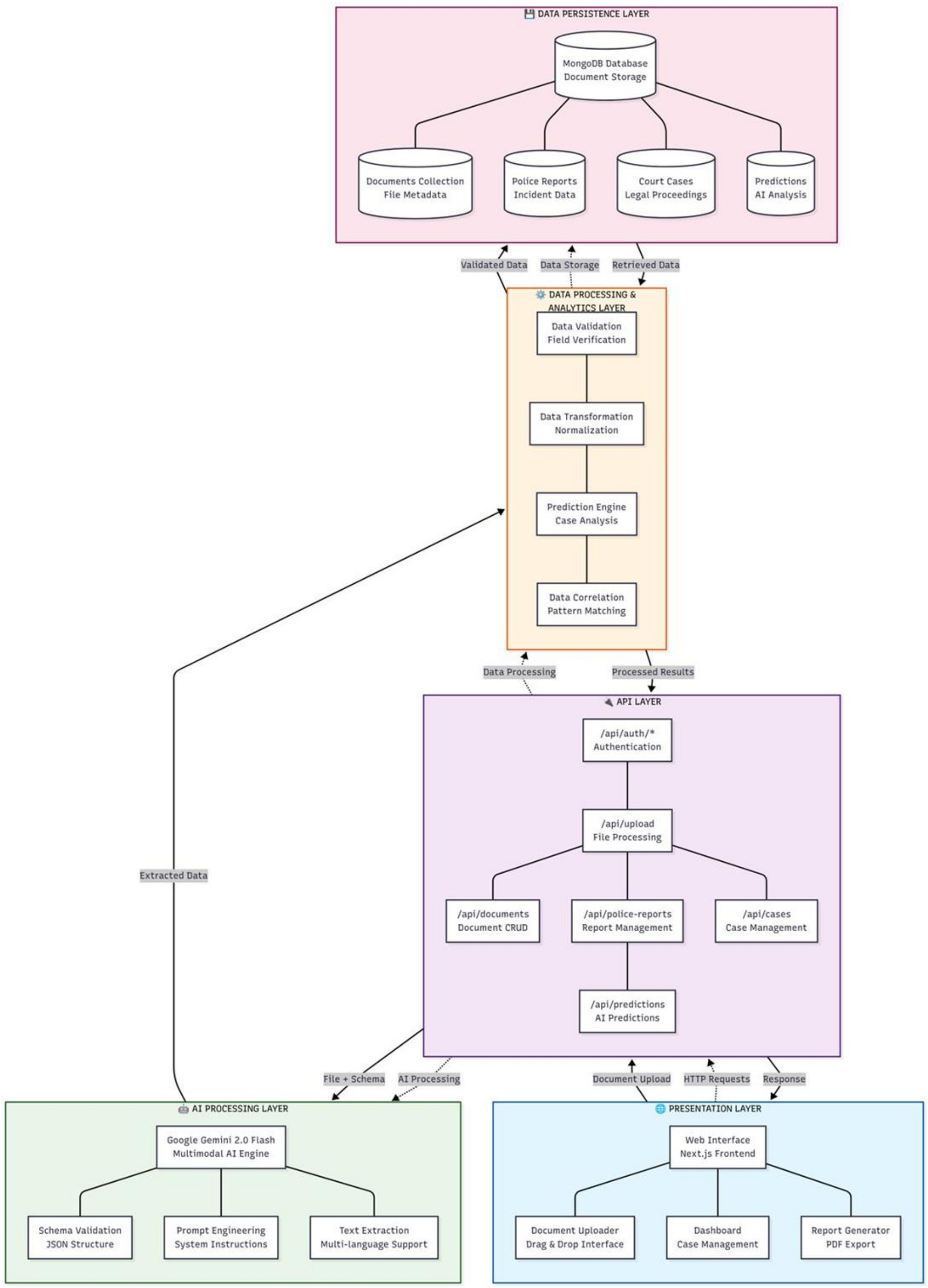


Figure 2 : Component Diagram for Automated Data Collection and Management

The component architecture for extracting data from police reports and court orders using AI-powered OCR with Gemini LLM follows a layered, modular design that ensures scalability, maintainability, and high accuracy in document processing. The system is built on a multi-tiered architecture that separates concerns and enables efficient processing of legal documents.

1. Input Layer

The Input Layer constitutes the primary interface for data acquisition and user engagement. This layer supports flexible ingestion methods, enabling the system to accommodate heterogeneous document formats and sources:

```
const policeReportSystemInstruction =
"You are an expert in extracting structured data from unstructured or semi-structured text in police reports. Given a file (which may be an image, PDF, DOCX, CSV,
"caseId (e.g., 'WCIB 367/04', 'BR 1082/25')\n" +
"incidentDate (e.g., 'February 27, 2025')\n" +
"incidentTime (e.g., '10:30 AM', '14:45', or 'Unknown')\n" +
"reportDate (e.g., 'March 1, 2025')\n" +
"victimName (e.g., 'Alina Oslopova' or leave 'Unknown' if redacted)\n" +
"victimAge (e.g., '25 years', or 'Unknown')\n" +
"victimGender (e.g., 'Female', 'Male', 'Unknown')\n" +
"victimNationality (e.g., 'Sri Lankan', 'Russian', or 'Unknown')\n" +
"perpetratorName (e.g., include if mentioned; else 'Unknown')\n" +
"perpetratorGender (e.g., 'Female', 'Male', 'Unknown')\n" +
"perpetratorNationality (e.g., 'Sri Lankan', 'Russian', or 'Unknown')\n" +
"relationshipToVictim (e.g., 'Spouse', 'Stranger', 'Friend', 'Employer', 'Unknown')\n" +
"incidentLocation (e.g., 'No. 39/3C, Kaluwella, Galle')\n" +
"incidentSummary (a detailed summary of the incident, including key events and context)\n" +
"typeOfViolence (e.g., 'Physical assault', 'Verbal threat', 'Sexual harassment', 'Unknown')\n" +
"injuryDescription (if any: e.g., 'swollen cheek', 'bleeding nose')\n" +
"evidenceMentioned (e.g., 'Medical Report', 'CCTV footage', 'Police Scene Report')\n" +
"reportedToAuthorities ('Yes', 'No')\n" +
"actionTaken (e.g., 'Victim taken to hospital', 'CCTV retrieved', 'Suspect summoned')\n" +
"recurrence (e.g., 'First-time incident', 'Repeat incident', 'Ongoing domestic violence')\n" +
"caseStatus (e.g., 'Ongoing', 'Referred to court', 'Concluded', 'Unknown')\n" +
"relevantLaws (e.g., 'Sri Lanka Penal Code 314, 316', or 'Children's Charter 1989')\n" +
"priorCriminalHistory (e.g., 'No prior record', 'Convicted of theft in 2020', or 'Unknown')\n" +
"The text may not follow a consistent format, and some fields may be missing, ambiguous, or in a non-English language (e.g., Sinhala).\n\n" +
"Guidelines:\n" +
"1. For caseId, look for identifiers like 'WCIB', 'MCR/', 'BR', or similar (e.g., 'WCIB 367/04', 'MCR/122/25'); if unclear, use 'Unknown'.\n" +
"2. For incidentDate and reportDate, accept various formats (e.g., 'February 27, 2025', '2025-02-27') and distinguish between them. Translate Sinhala dates to Eng
"3. For victimName, extract as provided (e.g., 'Alina Oslopova', '[illegible]'); if redacted or unclear, use 'Unknown'.\n" +
"4. For ages, extract numerical values or phrases like '25 years'; if not specified, use 'Unknown'.\n" +
"5. For gender, identify terms like 'Male', 'Female', or Sinhala equivalents (e.g., '[illegible]' for 'Male'); if unclear, use 'Unknown'.\n" +
"6. For victimNationality and perpetratorNationality, infer from context (e.g., 'Sri Lankan' from location or name); if unclear, use 'Unknown'.\n" +
"7. For relationshipToVictim, infer from context (e.g., 'Spouse' in domestic cases, 'Stranger' in theft); translate if in Sinhala; if unclear, use 'Unknown'.\n" +
```

Figure 3 : Prompt for data extraction

- **Document Upload Interface:** A Next.js-based web application forms the user-facing front end, featuring a drag-and-drop mechanism that supports diverse file types such as PDFs, Word documents (DOCX), scanned images (JPG, PNG), CSV, and plain text (TXT) files. The interface integrates real-time file validation, upload progress monitoring, and document categorization capabilities (distinguishing police reports from court orders), optimizing usability and accuracy in document submission.
- **Multi-Channel Data Acquisition:** Document ingestion occurs via direct web uploads and secure API endpoints, facilitating seamless integration with external law enforcement databases and legacy systems. This dual-path acceptance caters

both to digitized versions of case files and scanned hardcopies, ensuring comprehensive data coverage.

- **Security and Access Control:** Role-based authentication and authorization mechanisms are enforced through Next.js middleware, restricting document access and upload privileges to authorized personnel only. This layer ensures compliance with regulatory and privacy standards by safeguarding sensitive investigative data against unauthorized exposure.

2. Document Processing Layer

At the core of the system lies an advanced Document Processing Layer that harnesses Google Gemini 2.0 Flash's multimodal AI capabilities to perform intelligent document comprehension without relying on traditional OCR pipelines, thereby overcoming limitations of accuracy and preprocessing overhead:

```
const policeReportSchema = {
  type: "OBJECT",
  required: ["cases"],
  properties: {
    cases: {
      type: "ARRAY",
      items: {
        type: "OBJECT",
        required: [
          "caseId", "incidentDate", "reportDate", "victimName", "victimAge", "victimGender",
          "victimNationality", "perpetratorName", "perpetratorGender", "perpetratorNationality",
          "relationshipToVictim", "incidentLocation", "incidentSummary", "typeOfViolence",
          "injuryDescription", "evidenceMentioned", "reportedToAuthorities", "actionTaken",
          "recurrence", "caseStatus", "relevantLaws", "incidentTime", "priorCriminalHistory"
        ],
        properties: {
          caseId: { type: "string" },
          incidentDate: { type: "string" },
          reportDate: { type: "string" },
          victimName: { type: "string" },
          victimAge: { type: "string" },
          victimGender: { type: "string" },
          victimNationality: { type: "string" },
          perpetratorName: { type: "string" },
          perpetratorGender: { type: "string" },
          perpetratorNationality: { type: "string" },
          relationshipToVictim: { type: "string" },
          incidentLocation: { type: "string" },
          incidentSummary: { type: "string" },
          typeOfViolence: { type: "string" },
          injuryDescription: { type: "string" },
          evidenceMentioned: { type: "string" },
          reportedToAuthorities: { type: "string" },
          actionTaken: { type: "string" },
          recurrence: { type: "string" },
          caseStatus: { type: "string" },
          relevantLaws: { type: "string" },
          incidentTime: { type: "string" },
          priorCriminalHistory: { type: "string" },
        },
```

Figure 4 : Police report Data schema

- **Multi-Modal Document Processing:** This layer processes raw document inputs ,including images, scanned PDFs, and text files, by simultaneously performing text extraction, layout and structural analysis, handwritten text recognition, and multilingual interpretation (e.g., Sinhala to English translation). Additionally, it accurately extracts tabular and form-based data embedded within forensic documents.

- **Document Type Classification:** Leveraging content-based semantic cues, the system automatically categorizes documents into police reports and court orders, directing them through domain-specific analysis pipelines tailored to the unique structural and informational characteristics of each document type.
- **Real-Time Processing Pipeline:** Immediate document transformation is achieved through a streamlined pipeline that converts input files into Base64-encoded representations suitable for Gemini API processing. Extracted data is returned as structured JSON schema outputs, accompanied by metadata such as processing duration and system performance metrics.

3. AI Processing Layer

The AI Processing Layer is responsible for intelligent data extraction and contextual analysis utilizing sophisticated prompt engineering in conjunction with Google Gemini 2.0 Flash:

- **Model Configuration:** The Gemini model is finely tuned with domain-specific parameters (e.g., temperature of 0.5, topP of 1, topK of 1) to balance precision and diversity during text generation, optimizing for forensic accuracy and consistency.
- **System Instructions Engine:** A library of specialized prompts equips the AI to differentiate between document contexts, apply privacy-preserving transformations (e.g., anonymizing personal identifiable information), support multilingual inputs, and extract contextually relevant fields even when explicit data is missing.
- **Structured Output Schema:** Extraction results adhere to predefined JSON schemas that enforce uniformity, comprising over 24 fields for police reports (inclusive of case details, victim and perpetrator profiles, incident narratives) and 15 fields for court orders (capturing legal procedures, charges, verdicts, and sentencing information). This structured approach enables high fidelity in data representation and downstream processing.

4. Data Processing and Storage Layer

Following extraction, the system incorporates a robust Data Processing and Storage Layer to validate, normalize, and persist case information securely:

- **Data Transformation and Validation:** Extracted JSON data undergoes comprehensive parsing to ensure schema conformity, correct erroneous or incomplete values, execute cross-field consistency verifications, and introduce placeholder values ("Unknown") where data is legitimately missing, maintaining data integrity.
- **Database Operations:** Utilizing MongoDB coupled with Mongoose ODM, the system stores documents and their associated structured data as linked entities. Unique document identifiers (UUIDs) support traceability, while audit logs capture processing events and timing metrics to facilitate system monitoring and forensic auditing.
- **Data Persistence Strategy:** The repository maintains document metadata (file attributes, processing timestamps), structured case data, and version-controlled updates to support case lifecycle management and historical record preservation.

5. Intelligence and Analytics Layer

This advanced layer applies AI-driven analytics to enhance investigative decision-making and uncover latent case insights:

- **Predictive Analytics Integration:** Employing machine learning models on extracted data, the system forecasts case outcomes, assesses risk profiles, and evaluates probability metrics. Additionally, it identifies recurrent patterns and behavioral trends, thereby augmenting investigative heuristics.
- **Data Correlation Engine:** Sophisticated algorithms analyze temporal relationships across cases, map geographic proximities, detect recurring offenders or modus operandi, and establish cross-case linkages to guide comprehensive investigations.

6. Output and Presentation Layer

Finally, the system offers versatile interfaces and reporting capabilities tailored for investigative stakeholders:

- **User Dashboard Interface:** A responsive web dashboard supports real-time case monitoring, enriched by advanced search, filtering mechanisms, and document viewers that overlay extracted data for inline validation. Interactive visualizations relay case status, analytics, and pattern insights intuitively.

- **Report Generation System:** Employing js PDF and react-to-print libraries, the system generates customizable PDF reports aligned with organizational templates and user roles. These reports can be exported and integrated with external judicial or administrative systems to facilitate case documentation and presentation.
- **API Services Layer:** RESTful APIs provide controlled data access, enable real-time synchronization between subsystems, and manage authentication tokens for secure external integration, ensuring interoperability within broader law enforcement infrastructures.
- **System Security and Technical Considerations**

  End-to-end security underpins the architecture through rigorous API key management, input sanitization, rate limiting, secure file handling, and temporary storage strategies to prevent data leakage or denial-of-service vulnerabilities. The processing pipeline supports batch uploads, real-time progress tracking, automatic error recovery, and retry mechanisms to ensure operational resilience and scalability across diverse investigative workloads.

2.3.2. Multi-modal Behavioral and Emotional Analysis

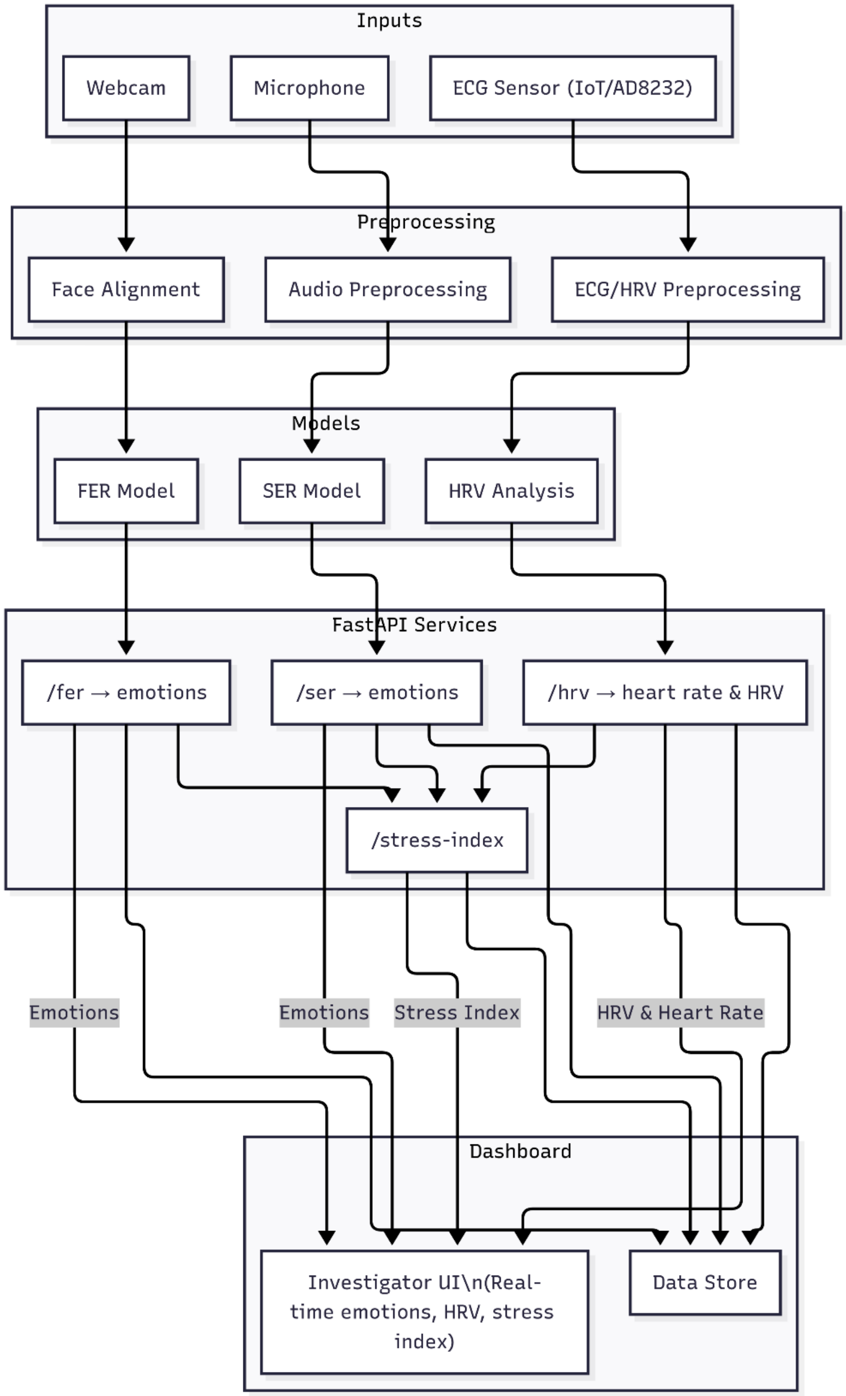


Figure 5 : Component Diagram Multimodal Behaviour& Emotional Analysis

The system design of the Multimodal Behavioural and Physiological Analysis (MBPA) module is centred on providing investigators with a real-time, evidence-based behavioural monitoring system that fuses multiple input modalities, facial expressions, speech signals, and physiological heart rate variability, into a unified Stress Index. The design was guided by four key principles: modularity, scalability, interpretability, and real-time responsiveness. By adopting a microservices-based architecture using FastAPI servers, each unimodal model operates as an independent service that communicates with the central dashboard through REST APIs. This approach ensures that models can be trained, deployed, and scaled independently, while the dashboard functions as the single point of interaction for investigators.

The MBPA system is designed as a five-layer architecture, consisting of:

1. Input Acquisition Layer
2. Preprocessing Layer
3. Unimodal Analysis Layer
4. Multimodal Fusion Layer
5. Visualization and Dashboard Layer

Each layer is described in detail below, along with the rationale for its design and integration into the overall architecture.

- **Input Acquisition Layer**

The first stage of MBPA involves capturing multimodal signals from suspects during interrogation sessions.

- Facial data is obtained from a standard webcam operating at 25–30 frames per second. This choice of hardware emphasizes cost-effectiveness and accessibility, as webcams are already present in most interrogation environments. Unlike laboratory-grade cameras, webcams provide realistic video quality under challenging conditions such as varying lighting, which helps ensure the robustness of the FER system.
- Speech data is captured through a microphone, enhanced with Voice Activity Detection (VAD) to filter out silence and irrelevant background noise. Audio is sampled at 22.05 kHz and segmented into short frames of 2–3 seconds for analysis.
- Physiological data is obtained using the AD8232 single-lead ECG sensor, attached to the suspect via electrodes. This sensor was chosen for its compact size, affordability, and reliability in capturing ECG waveforms. To improve

training and validation, additional ECG and GSR signals were integrated from a publicly available dataset (YAAD), ensuring diversity and coverage of different affective states.

All inputs are time-stamped at acquisition, ensuring that they can later be aligned for multimodal fusion. To maximize robustness, the acquisition layer also computes signal quality indicators: FER tracks face detection confidence; SER estimates audio quality based on SNR and speech probability; and HRV tracks ECG artifact rates. These indicators are passed downstream to influence adaptive fusion weights.

- **Preprocessing Layer**

Raw multimodal inputs often contain noise and inconsistencies that can severely degrade model performance. Therefore, preprocessing pipelines were designed for each modality to clean, normalize, and standardize inputs:

  - Facial data preprocessing: Grayscale conversion and histogram normalization reduce illumination variability. Face detection and alignment are performed using Dlib's landmark predictor to standardize head pose. During training, data augmentation methods (rotation, flipping, brightness shifts) were applied to enhance robustness against occlusion and pose variation.
  - Speech data preprocessing: Noise suppression and silence trimming remove irrelevant signals. Librosa was used to extract features such as MFCCs, chroma, spectral contrast, and prosodic cues. Data augmentation was employed to simulate real-world conditions, including pitch shifting, time-stretching, and noise injection.
  - ECG preprocessing: ECG signals were filtered using band-pass filters to eliminate baseline drift and high-frequency noise. RR intervals were extracted to compute HRV features such as RMSSD, SDNN, LF/HF ratio, and Baevsky's Stress Index. Baseline ECGs were collected at the start of sessions to normalize individual variability.

This layer ensures that heterogeneous, noisy raw data is transformed into consistent, high-quality features suitable for model inference.

- **Unimodal Analysis Layer**

The unimodal analysis layer processes each modality using dedicated models, each deployed as an independent FastAPI microservice.

  - Facial Emotion Recognition (FER): A CNN-based deep learning model trained on

FER-2013 classifies seven emotions (angry, disgust, fear, happy, neutral, sad, surprise). The model processes aligned face chips and produces emotion probabilities at ~1 Hz (every 0.5–1 second). FER is deployed as a FastAPI server: the dashboard sends cropped face images via REST API, and the server responds with JSON-formatted probabilities.

- Speech Emotion Recognition (SER): A CNN–BiLSTM hybrid with attention trained on RAVDESS predicts eight emotions (neutral, calm, happy, sad, angry, fear, disgust, surprise). Audio features are passed to the SER FastAPI server, which outputs posterior probabilities every 2–3 seconds.
- Heart Rate Variability (HRV): Time- and frequency-domain HRV features are computed from ECG windows of 60–300 seconds. The HRV FastAPI server accepts RR interval data and returns normalized HRV metrics along with a physiological stress subscore (S_hrv).

Each model is encapsulated in its own FastAPI server, enabling loose coupling. This design ensures that any unimodal service can be retrained, upgraded, or replaced without affecting the rest of the pipeline.

- **Multimodal Fusion Layer**

The fusion layer integrates unimodal outputs into a single Stress Index (0–100), providing a dynamic measure of stress.

- Emotion-driven Stress (S_emo): FER and SER outputs are mapped to the Valence–Arousal (V–A) space, following the circumplex model of affect. Posterior probabilities of discrete emotions are transformed into expected valence and arousal values, then combined into an emotion-based stress score, where high arousal and negative valence yield higher stress.
- Physiological Stress (S_hrv): Derived from normalized HRV features such as RMSSD, SDNN, LF/HF, and Baevsky's Stress Index. These are combined into a subscore using a weighted linear model and logistic squashing.

- Adaptive Fusion: The final Stress Index is computed as:

$$\mathbf{S=\alpha\cdot Shrv+\beta\cdot Semo,\alpha+\beta=1}$$

Weights α and β are dynamically assigned based on signal quality indicators (e.g., if the face is occluded, α increases; if ECG is noisy, β increases). Temporal smoothing is applied using an exponential moving average (EMA) to ensure stability, while still retaining responsiveness to sudden changes.

The fusion logic itself is deployed as a FastAPI backend service, which queries each unimodal model's API, performs synchronization, and outputs the fused Stress Index to the dashboard.

- **Visualization and Dashboard Layer**

The dashboard is the central investigator interface that visualizes multimodal outputs in real time. It is implemented as a web-based application that continuously queries the FastAPI services for new results. Key features include:

  - Live video feed with overlaid FER predictions (emotion labels + confidence scores).
  - Dynamic plots of SER trajectories, showing vocal stress variations over time.
  - ECG and HRV graphs displaying heart rate changes and HRV indices.
  - A continuous Stress Index (0–100) trendline, updated every 5–10 seconds, with clear indicators of modality contributions (e.g., 65% HRV, 35% FER/SER).
  - Alerts that trigger when stress crosses predefined thresholds.

The dashboard prioritizes clarity and interpretability, ensuring that investigators are not burdened with raw data but instead receive clear, actionable insights. It functions strictly as a decision-support tool, not as a replacement for investigator judgment.

- **System Integration and Scalability**

The use of FastAPI microservices enables MBPA to be deployed in a modular, scalable, and secure fashion. Each service runs inside a container (Docker), which can be replicated or scaled independently. This ensures:

  - **Scalability**: Multiple interrogation rooms can be supported by deploying multiple containers, each serving FER, SER, and HRV models.
  - **Fault Tolerance**: If one service fails (e.g., SER during background noise), the dashboard continues using the remaining modalities.
  - **Interoperability**: REST APIs allow MBPA to integrate seamlessly with other modules in the interrogation platform, such as dynamic question generation and testimony correlation.
  - **Security**: API-level authentication, encrypted communication (HTTPS), and access control safeguard sensitive data.

This architecture ensures that MBPA can be deployed both locally (e.g., in a police interrogation room) and remotely (e.g., via cloud servers for centralized monitoring). Its modular design also allows for easy future extensions, such as adding gaze tracking or galvanic skin response (GSR) analysis, by simply introducing additional FastAPI services.

The system design of MBPA provides a comprehensive framework for multimodal behavioural monitoring in criminal interrogations. By structuring the pipeline into acquisition, preprocessing, unimodal analysis, fusion, and visualization, the design ensures both technical robustness and operational usability. Deploying each model as a FastAPI microservice enables modularity, scalability, and fault tolerance, while the centralized dashboard presents a unified Stress Index to investigators in real time. The system not only leverages advanced deep learning and signal processing techniques but also prioritizes transparency, adaptability, and security, bridging the gap between academic research and practical deployment in law enforcement contexts.

2.3.3. Dynamic Question Generation and Response Analysis

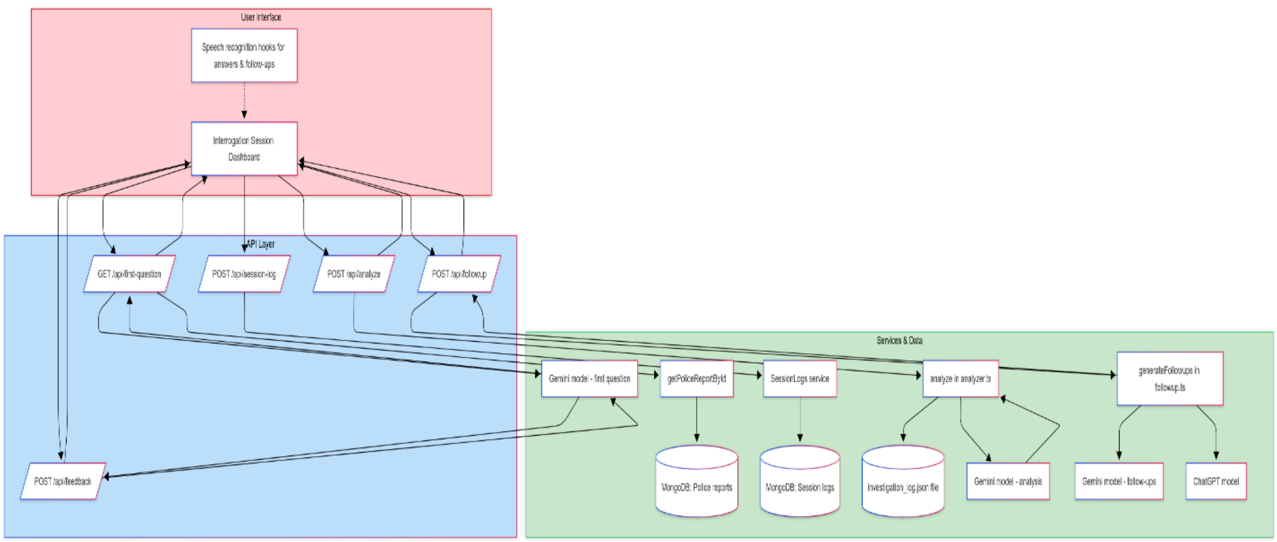


Figure 6: System Overview Diagram

The dynamic question-generation and response analysis system comprises a modern full-stack application built around a React/Next.js front-end, a set of server-side API endpoints, a persistence layer backed by MongoDB, and integrations with large language models such as Google's Gemini and OpenAI's GPT-4. The following section explains the major components and their interactions, justifying design decisions with references to external sources.

1. **Frontend (Interrogation Session Dashboard)**

The investigator interface is a Next.js client component that presents the interrogation dashboard. It displays the suspect's case details, controls the session lifecycle (start, pause/resume, stop), shows a live timer, and renders prompts, answers and analytical feedback. State management is handled with React hooks and the UI is styled with Tailwind CSS components. Voice input is enabled via a custom useSpeechRecognition hook, which wraps the browser's Web Speech API. This abstraction allows investigators to record spoken answers and manually dictate follow-up questions, improving ease of use in field settings.

All network interactions occur through fetch calls to api endpoints. The front-end never directly communicates with the database or LLMs; instead, it relies on typed JSON responses returned by the server. This separation of concerns keeps the client lightweight and secure.

2. **API layer (Next.js API routes)**

- **first-question (GET)** – Generates the initial open-ended question. The route uses getPoliceReportById() to fetch the case record from MongoDB and formats a detailed prompt describing the incident, victim, perpetrator and other contextual fields. It then calls the Gemini API via Google's @google/generative-ai client with the model gemini-2.0-flash. The Gemini API accepts a model name and contents array and returns text output, which the handler extracts and sends back to the front-end as the initial question.
- **analyze (POST)** – Analyses the suspect's answer. The handler invokes the analyze() function. This function constructs a structured prompt instructing the model to rate relevance (0–10), list facts gathered, highlight missing information and suggest points for deeper probing. The same Gemini model is called to generate the analysis. The raw LLM output is parsed into structured fields (relevanceScore, dataGathered, missingData and pointsToExplore) and appended to an investigation_log.json file for auditing.
- **followup (POST)** – Generates follow-up questions. The service calls two generative models: Google's Gemini and OpenAI's GPT-4. The Gemini call uses the same client library as above, while the ChatGPT call uses the official openai Node SDK. As shown in a tutorial on building a CLI chatbot with the ChatGPT API, the OpenAI client is created with an API key and the method is

invoked with a list of messages (system prompt and user prompt) and the desired model. The follow-up handler returns one question from each model and indicates the preferred model, allowing the investigator to choose which LLM to trust in subsequent steps.

- **feedback (POST)** – Accepts the investigator's selection (either a generated follow-up or a manually typed question) and returns the next primary question.
- **session-log (POST)** – Persists the session. Before concluding the session, the client sends an array of entries (questionId, question, answer and numeric relevance score) along with a sessionId and investigationId. This endpoint interacts with the SessionModel through Mongoose. The service uses findOneAndUpdate() with upsert: true to perform atomic updates: if a session log for the given sessionId exists, it updates the document; otherwise it inserts a new document. This ensures that concurrent updates to the same session are consistent, and no duplicates occur.

3. **Services and persistence layer**

Police report service

The function abstracts database access for retrieving case summaries. It uses Mongoose models to query the PoliceReports collection in MongoDB. By encapsulating data retrieval in a service, the API handler remains focused on prompt construction and result formatting.

Session log service

The session log service wraps Mongoose operations for creating, updating, querying and deleting Session documents. It uses ensureObjectId() to validate that incoming IDs are valid MongoDB ObjectIds. At the end of a session, the service writes the session log to the database along with timestamps and optional caseIds. This persistent record supports auditing and analysis of interrogation strategies.

In-memory and file logging

Besides database storage, the analyzer module writes each analysis entry as a JSON line to database. This file serves as an immediate audit trail and can be imported into analytics tools for offline review.

LLM integration

The core intelligence of the system relies on two families of large language models:

1. **Google Gemini (Generative AI)** – The Gemini API supports multi-modal input but is used here purely for text generation. The system leverages this for first-question generation, answer analysis and follow-up question generation. Using a consistent model ensures that the follow-up questions remain aligned with the initial interviewing style.
2. **OpenAI GPT-4 via Chat Completions** – For alternative follow-ups, the system uses OpenAI's GPT-4 through the Node SDK. The message structure includes roles for "system" instructions, "user" questions and "assistant" responses, enabling context-aware conversation. This design allows the investigator to compare the questioning styles of two different LLMs and select the most appropriate follow-up.

2.3.4. Incident Correlation and Pattern Identification

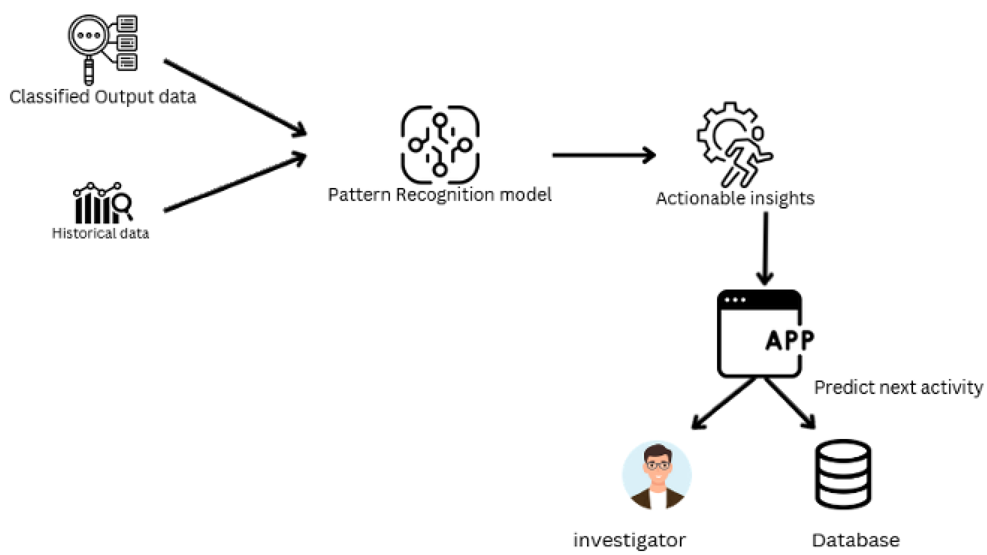


Figure 7: Component Diagram for Incident Correlation and Pattern Identification

The illustrated component represents the **pattern recognition and decision-support layer** of the proposed investigative system. It is responsible for transforming raw and historical case data into actionable insights that can guide investigators in real time.

The process begins with the ingestion of **classified output data** from the machine learning model alongside **historical case records** stored in the database. The classified outputs include predictions generated by the Random Forest classifier, such as the likelihood of criminal responsibility based on structured attributes (victim age, perpetrator age, gender, type of violence, and evidence mentions). Historical data, on the other hand, provides a contextual basis, allowing the system to compare current cases against past incidents to identify recurring patterns or anomalies.

These two streams of information are passed into the **Pattern Recognition Model**, which analyzes them to uncover hidden relationships. By correlating features across multiple cases, the model can detect trends such as repeated offender behavior, similarities in modus operandi, or recurring risk factors associated with victim vulnerability. This pattern recognition step is crucial in enabling investigators to move beyond isolated case analysis and instead evaluate broader investigative connections.

The outputs of this analysis are then translated into **actionable insights**. Rather than presenting raw probabilities or isolated predictions, the system synthesizes its findings into meaningful recommendations for investigators. For example, the system may highlight that a current case shares significant similarities with prior high-risk incidents or that the victim demonstrates critical vulnerability factors.

These insights are delivered through the **application interface (App)**, which acts as the central platform for investigator interaction. Within the app, the system not only presents current findings but also **predicts potential next activities**. This predictive capability is valuable in anticipating possible escalation scenarios or linking current incidents to potential future outcomes.

Finally, the insights are made accessible to both the **investigator** and the **database**. For the investigator, the app provides an interactive dashboard that enables real-time decision support during ongoing cases. For the database, the system continuously updates stored records with the latest case outputs and correlations, ensuring that future analyses benefit from accumulated knowledge.

2.4. System Flow & Implementation

2.4.1. Automated Data Collection and Management

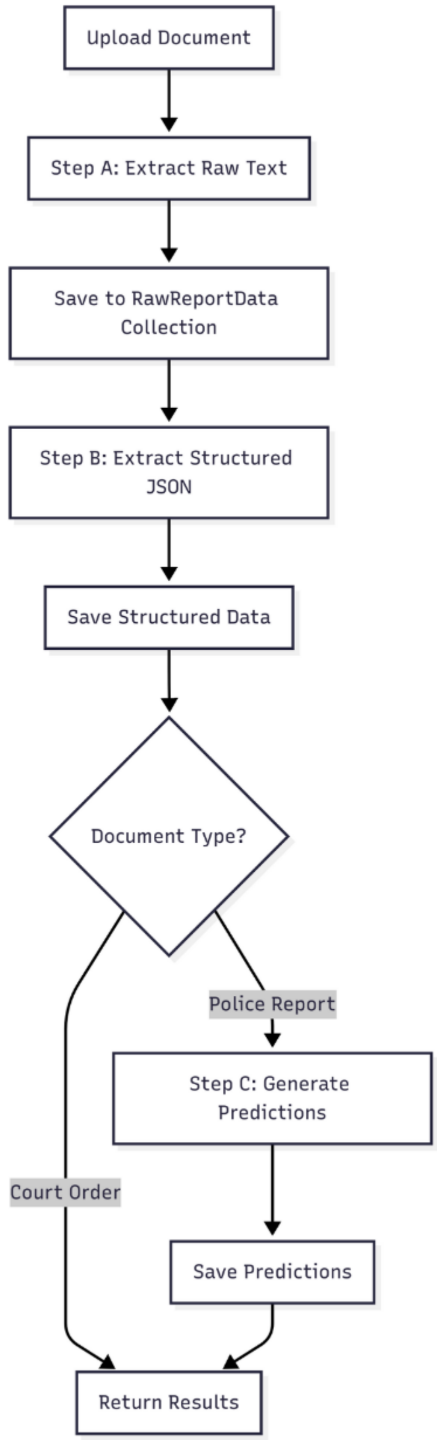


Figure 8 : Chain Prompt

The implementation of the proposed system was carried out in several phases to ensure that both functional and non-functional requirements were met. The system provides a secure environment for investigators to register, manage case data, and analyze information extracted from police reports and related documents.

- **User Registration and Access Control**

  User registration begins with the creation of an account within the system. To safeguard against unauthorized access, each registration request undergoes verification by an administrative officer, who validates the identity of the applicant before granting access. This process ensures that only legitimate law enforcement personnel are able to utilize the system, thereby reducing the risk of external parties gaining entry.

- **People Registry Database**

  During an investigation, details of the involved parties are stored in a dedicated People Registry Database. Information recorded includes the individual's full name, date of birth, gender, national identity card number, address, and a photographic record. This structured repository provides investigators with a centralized source of personal data relevant to ongoing and historical cases, ensuring consistency and accessibility throughout the investigation process.

- **Investigator Dashboard**

  Upon successful login, users are presented with the Home Dashboard, which serves as the central hub of the application. This interface allows investigators to view Recent Investigations and Active Investigations, along with corresponding progress levels and due dates. The Investigations Page provides access to both ongoing and archived cases, while the Learn Page delivers global crime updates and interactive tutorials designed to enhance investigators' knowledge and awareness of international practices.

- **Document Upload and Data Extraction**

  The system enables investigators to upload documents by referencing a unique Report ID. Users are prompted to specify the type of document being uploaded, such as a Police Report or a Court Order. Once confirmed, the system automatically processes the file, extracting relevant data and converting it from unstructured text into structured formats.

The extracted data is organized within the Report Details section. This is further divided into categories:

1. **Report Information**: Case ID, incident date and time, report date, case status, recurrence, and relevant legal provisions.
2. **Actions and Authorities**: Actions taken, reporting details to relevant authorities, and the associated Document ID.
3. **Incident Evidence**: Divided into *Incident Details* (location, type of violence, and summary) and *Injuries and Evidence* (injury descriptions and references to material evidence).
4. **Involved Parties**: Records the name, age, gender, nationality, relationship to victim, and any prior criminal history of victims and perpetrators.
5. **Documents**: Provides a preview of the original physical report, alongside the ability for users to make manual edits if necessary.

The system further allows investigators to generate and download structured reports in a printable PDF format. Case Summaries, including AI-generated insights and predictions derived from extracted data, are also available within this section.

- **Report Management and Analytics**

  Processed police reports can be revisited through the Recent Documents tab, while the Processing Stats module offers a statistical overview of system performance. This includes data on average processing times and the total number of documents processed across daily, monthly, and yearly periods. Such analytics enable investigators and administrators to monitor system usage and efficiency.

- **Case Creation and Management**

  A new investigation can be created within the system by recording key metadata such as the case title, case type, priority level, involved parties, and any associated reports. This functionality ensures that case files remain structured, easily accessible, and dynamically linked to the documents and data stored within the system.

2.4.2. Multimodal Behavioral and Emotional Analysis

The implementation phase of the Multimodal Behavioural and Physiological Analysis (MBPA) module translated the system design into a functional prototype that could operate in real time within an interrogation support environment. This chapter details the development environment, preprocessing pipelines, model training and optimization, deployment strategy using FastAPI microservices, integration with the investigator dashboard, and the technologies that enabled the system. Particular emphasis was placed on ensuring that the final implementation was modular, scalable, and robust enough to handle real-world variability in input signals.

Development Environment

Implementation was carried out primarily in Python 3.10, chosen for its versatility and extensive ecosystem of libraries for deep learning, signal processing, and system integration. Development was conducted on a laptop equipped with an NVIDIA RTX 4070 GPU (8 GB VRAM) and CUDA 12.x acceleration. This hardware significantly reduced model training time and enabled real-time inference during deployment.

Version control and collaboration were managed using GitHub, ensuring consistent tracking of code revisions, issue management, and backup. Containerization was supported with Docker, allowing services to be packaged with all dependencies and deployed consistently across environments.

Data Preprocessing Pipelines

Preprocessing was critical to prepare multimodal data for model training and inference. Separate pipelines were developed for facial images, speech signals, and ECG recordings:

- **Facial Data**: Frames captured from webcam feeds were preprocessed using OpenCV and Dlib. Each frame was converted to grayscale, normalized to reduce illumination effects, and cropped to 48×48 pixels. Dlib's landmark predictor ensured face alignment. During training, augmentation techniques such as horizontal flipping, rotation, and brightness adjustments improved generalization.

- **Speech Data**: Audio signals were processed with Librosa, which extracted MFCCs, chroma features, spectral contrast, and delta features. Preprocessing included trimming silence, denoising, and segmenting into 2–3 second windows.

Augmentation simulated real-world variability using pitch shifting, time-stretching, and additive noise.

- **ECG Data**: Physiological signals were filtered using a Butterworth band-pass filter to remove baseline wander and high-frequency noise. R-peaks were detected and RR intervals extracted. HRV features including RMSSD, SDNN, LF/HF ratio, and Baevsky's Stress Index were computed on rolling windows of 60–300 seconds. Each subject's baseline was recorded at rest to normalize HRV measures.

These preprocessing steps ensured that inputs were consistent, noise-resilient, and suitable for downstream model inference.

Model Training and Optimization

Each modality was modeled independently before integration into the MBPA system:

- **Facial Emotion Recognition (FER)**: A convolutional neural network (CNN) was implemented in PyTorch, trained on the FER-2013 dataset. Training used the Adam optimizer with learning rate scheduling, batch size 64, and early stopping to avoid overfitting. The model achieved convergence after 39 epochs with test accuracy of ~63%.
- **Speech Emotion Recognition (SER)**: A hybrid CNN–BiLSTM with attention was implemented in PyTorch and trained on the RAVDESS dataset. Acoustic features extracted by Librosa were passed to the model, which captured both spectral and temporal dependencies. Weighted cross-entropy was used to mitigate class imbalance. The model reached ~72% validation accuracy after ~160 epochs.
- **Heart Rate Variability (HRV)**: Instead of deep learning, HRV analysis relied on statistical and spectral features computed from ECG signals. Features were normalized against baseline and combined into a physiological stress subscore.

Training outputs included loss/accuracy curves, confusion matrices, and classification reports, which were later presented in the Results chapter for evaluation.

API Deployment with FastAPI

To enable modularity and scalability, each unimodal model was deployed as a standalone FastAPI microservice. This architecture allowed the dashboard to communicate with models through REST APIs, ensuring low-latency responses suitable for real-time interrogation support.

- **FER Service**: Accepts POST requests containing preprocessed face images and returns JSON-formatted emotion probabilities.
- **SER Service**: Accepts audio features and returns posterior probabilities across eight emotions.
- **HRV Service**: Accepts RR interval data, computes HRV indices, and returns both features and a physiological stress subscore.
- **Fusion Service**: Queries all unimodal services, aligns predictions using timestamps, and computes the unified Stress Index.

Each service was containerized with Docker to guarantee consistency across machines. This microservices approach ensured that models could be retrained or replaced independently without affecting the rest of the system.

Dashboard Integration

The investigator-facing dashboard was implemented using Next.js with TypeScript and styled using Tailwind CSS. This technology stack enabled a modern, component-based interface optimized for real-time responsiveness.

The dashboard queries the FastAPI services asynchronously and visualizes results using Plotly and Chart.js. It presents:

- Live video with overlaid facial emotion labels from the FER model.
- Real-time plots of vocal stress trajectories from the SER model.
- HRV graphs showing physiological variability and ECG-based stress markers.
- A continuous Stress Index (0–100) with modality contribution breakdowns.
- Alerts when stress levels exceed pre-defined thresholds.

Processed data and session logs are stored in MongoDB, chosen for its schema-less flexibility in handling multimodal JSON data. This allows storage of diverse records including raw predictions, stress trajectories, and dashboard events.

Challenges and Solutions

Several technical and operational challenges arose during implementation:

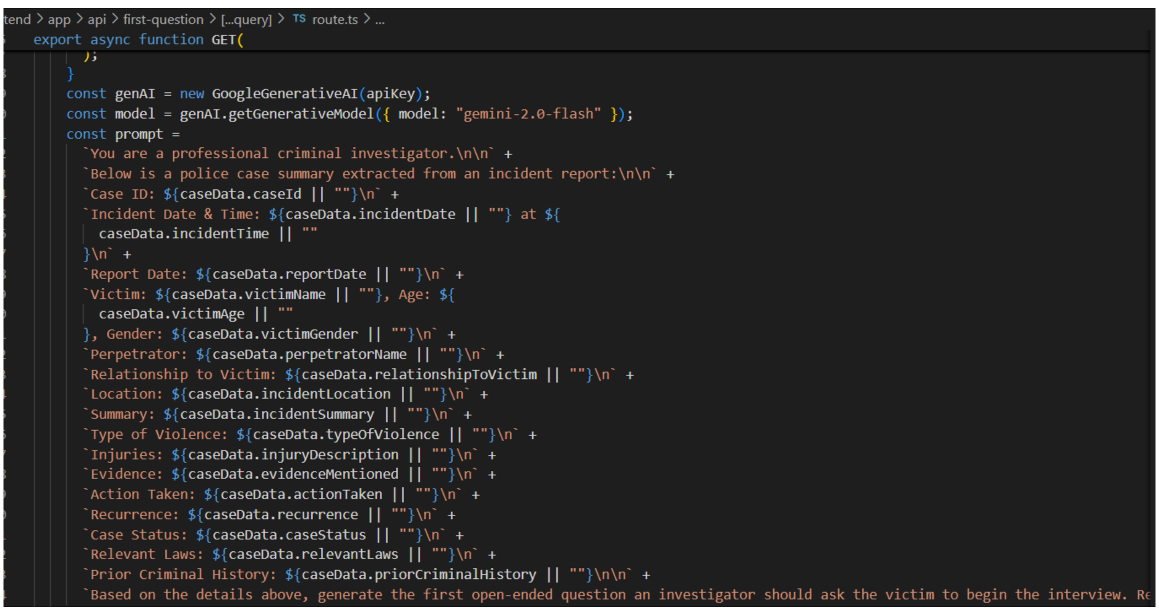
- **Data Imbalance**: FER-2013 and RAVDESS datasets exhibited imbalance across emotion classes. This was mitigated using class weighting and augmentation.

- **ECG Noise**: ECG signals were prone to artifacts caused by movement. Band-pass filtering and rejection of windows with >5% corrected beats addressed this issue.
- **Latency in Fusion**: Querying multiple APIs introduced latency. Asynchronous request handling in FastAPI reduced delays, and predictions were buffered for smooth dashboard updates.
- **Synchronization Across Modalities**: FER, SER, and HRV operate at different time scales. Timestamp alignment ensured that fusion combined temporally corresponding predictions.

These solutions strengthened the reliability and responsiveness of MBPA under realistic conditions.

2.4.3. Dynamic Question Generation and Response Analysis

The proposed system has been designed to replicate the natural flow of an interrogation session while embedding automation, intelligence, and adaptive questioning capabilities. Each session progresses through a series of structured stages, ensuring seamless interaction between the investigator and the AI system. The workflow is as follows:



```
tend > app > api > first-question > [...query] > TS route.ts > ...
  export async function GET(
      );
    }
    const genAI = new GoogleGenerativeAI(apiKey);
    const model = genAI.getGenerativeModel({ model: "gemini-2.0-flash" });
    const prompt =
      `You are a professional criminal investigator.\n\n` +
      `Below is a police case summary extracted from an incident report:\n\n` +
      `Case ID: ${caseData.caseId || ""}\n` +
      `Incident Date & Time: ${caseData.incidentDate || ""} at ${
        caseData.incidentTime || ""
      }\n` +
      `Report Date: ${caseData.reportDate || ""}\n` +
      `Victim: ${caseData.victimName || ""}, Age: ${
        caseData.victimAge || ""
      }, Gender: ${caseData.victimGender || ""}\n` +
      `Perpetrator: ${caseData.perpetratorName || ""}\n` +
      `Relationship to Victim: ${caseData.relationshipToVictim || ""}\n` +
      `Location: ${caseData.incidentLocation || ""}\n` +
      `Summary: ${caseData.incidentSummary || ""}\n` +
      `Type of Violence: ${caseData.typeOfViolence || ""}\n` +
      `Injuries: ${caseData.injuryDescription || ""}\n` +
      `Evidence: ${caseData.evidenceMentioned || ""}\n` +
      `Action Taken: ${caseData.actionTaken || ""}\n` +
      `Recurrence: ${caseData.recurrence || ""}\n` +
      `Case Status: ${caseData.caseStatus || ""}\n` +
      `Relevant Laws: ${caseData.relevantLaws || ""}\n` +
      `Prior Criminal History: ${caseData.priorCriminalHistory || ""}\n\n` +
      `Based on the details above, generate the first open-ended question an investigator should ask the victim to begin the interview. Re
```

Figure 9 : Data Extraction

1. **Start Session**
   The interrogation process begins when the investigator initiates the session by clicking the “Start Session” button on the interface. This action triggers a request to the first-question API endpoint, passing the unique case ID and the role of the

individual under interrogation (victim or perpetrator). The API handler retrieves the corresponding police report from the database and constructs a rich context prompt, embedding details such as incident summaries, entities, and prior testimonies. This context is then sent to Gemini, which generates an opening question tailored to the case background. The resulting question is immediately displayed on the investigator's interface, ensuring the session begins with a focused, evidence-driven prompt. Capture answer – The suspect responds verbally. The useSpeechRecognition hook transcribes the speech and populates the answer state. React state updates propagate through the UI.

2. **Capture Answer**
   Once the suspect responds, the system employs the useSpeechRecognition hook to capture the verbal response in real time. The audio is transcribed into text and stored within the React state, which continuously updates the user interface to display the suspect's answer. This transcription serves two purposes:
   i. maintaining an immediate record of the suspect's statement, and
   ii. preparing structured input for subsequent AI analysis. This feature significantly reduces reliance on manual notetaking, ensuring accuracy and minimizing transcription delays during the interrogation
3. **Analyze Answer**
   At this stage, the investigator triggers an analysis of the captured response by calling the analyze API endpoint. The current question–answer pair is transmitted to the backend server, which then queries Gemini. The model evaluates the answer for relevance, coherence, and completeness, while also extracting key data points (e.g., names, dates, locations, and described events). Additionally, the analysis highlights missing or ambiguous information and suggests lines of inquiry for further exploration. The structured output is then returned to the dashboard, where it is visualized by the investigator.
4. **Generate Follow-ups**
   To deepen the interrogation, the investigator clicks the "Generate Follow-ups" button. This action sends the Q&A pair to the followup API endpoint, which queries both LLMs in parallel. Each model generates one candidate follow-up question, resulting in at least two distinct suggestions. Alongside the questions, the system provides an indication of which model's response is likely more

contextually relevant, based on internal evaluation metrics. The investigator is then given flexibility: they may select one of the suggested follow-ups or alternatively formulate a manual question. This design ensures that AI recommendations assist investigators without removing human control from the interrogation process.

5. **Submit Feedback**
   Once a follow-up question is chosen, it is sent to the feedback endpoint. The backend processes the selection, updating the reinforcement component of the system. Positive selections reinforce the generation patterns of the chosen model, while manual overrides provide feedback on model limitations. The selected question is also used by the corresponding LLM to formulate the next primary interrogation prompt**,** effectively closing the loop and preparing the system for the subsequent cycle of questioning. This continuous loop establishes a reinforcement-based improvement mechanism, where investigators input directly refines future question generation.
6. **Stop Session**
   When the interrogation concludes, the investigator terminates the session by clicking the "Stop Session" button. The client calls the session-log APIendpoint, sending the complete record of session entries. The server commits this data to MongoDB using the findOneAndUpdate() method with the upsert option, ensuring atomic updates that either create a new record or update an existing one. This guarantees that the session log is securely stored, audit-ready, and retrievable for future reference, case correlation, or judicial review.

2.4.4. Incident Correlation and Pattern Identification

Input Case Data

The process begins when the investigator inputs a case ID into the dashboard interface. This action triggers an API request to the backend, where the system retrieves the structured case record from the MongoDB database. The retrieved data includes attributes such as victim age, perpetrator age, gender, type of violence, relationship status, and evidence mentions. Before further processing, the feature extraction module standardizes these attributes into numerical and categorical values, ensuring compatibility with the machine learning model.

Criminal Likelihood Prediction

Once the features are prepared, they are passed into the trained Random Forest classifier. The model analyzes the attributes and outputs a probability score representing the likelihood of criminal responsibility. This prediction is expressed as a percentage, giving investigators a quantifiable measure of case severity and offender risk. The result is then stored temporarily in the backend and sent forward for further assessment.

Victim Risk Assessment

In parallel with offender prediction, the system evaluates victim vulnerability using a rule-based risk scoring module. This component considers critical factors such as recurrence of violence, severity of injuries, relationship dynamics, and age of the victim. Each factor contributes to a weighted risk score, which is categorized as Low, Moderate, High, or Critical. This ensures the system balances offender profiling with victim protection.

Similarity and Correlation Analysis

To provide context, the system compares the current case with historical records. A BERT-based semantic similarity module encodes case summaries into embeddings and calculates cosine similarity with stored testimonies, identifying semantically related cases. In addition, a keyword correlation engine detects lexical overlaps such as repeated names, places, or weapons. Together, these analyses highlight potential connections that may not be obvious through manual review.

Verdict Generation

The verdict logic layer integrates outputs from the Random Forest prediction, victim risk assessment, and correlation modules. Based on predefined rules and thresholds, it classifies the case into one of three categories: Likely Criminal, Likely Victim, or Inconclusive. This ensures that multiple perspectives are considered before presenting a final decision to the investigator.

Explainability and Visualization

Finally, the explainability module produces a human-readable justification for the system's outputs. This includes listing the most influential features (such as evidence presence or age difference), displaying top similar cases from the similarity module, and highlighting correlated keywords. The complete result is then displayed in the investigator's dashboard in real time, with clear visualization of the likelihood score, victim risk badge, final verdict, and supporting explanations.

2.5. Technology Used

The integrated AI-Assisted Criminal Investigations System was developed using a diverse and complementary technology stack, carefully chosen to support large-scale testimony analysis, multimodal behavioural monitoring, case correlation, and dynamic interrogation support. Technology can be grouped into the following categories:

1. Programming Languages & Frameworks

- Python: Core language for AI and machine learning components, including data preprocessing, NLP pipelines (spaCy, NLTK, Pandas), statistical modeling (Scikit-learn), and deep learning frameworks (PyTorch).
- JavaScript/TypeScript: Used extensively for frontend and backend integration. Next.js with React was adopted across all modules for building responsive, investigator-friendly dashboards, while TypeScript provided type safety and reduced runtime errors in API-driven workflows.

2. Machine Learning & AI Models

- **Large Language Models (LLMs)**:
  - Google Gemini 2.0 for OCR-free multimodal document processing, dynamic question generation, and semantic analysis of testimonies.
  - OpenAI GPT-4 for generating alternative follow-up questions during interrogations, complementing Gemini.
  - Sentence Transformers (SBERT, BERT variants) for semantic similarity analysis of testimonies.
- **Deep Learning Models**:
  - Convolutional Neural Networks (CNNs) for Facial Emotion Recognition (FER) using FER-2013.
  - CNN–BiLSTM with Attention for Speech Emotion Recognition (SER) using RAVDESS.
  - HRV Analysis (ECG with AD8232 sensor) processed with NeuroKit2 and SciPy to extract physiological stress indicators.
  - Random Forest Classifier for structured case-based offender prediction.

3. Databases & Storage

- MongoDB / MongoDB Atlas: NoSQL databases were selected for storing structured/unstructured case records, testimony logs, physiological signals, and interrogation session histories.
- Google Cloud Storage: Used for secure cloud hosting, backup, and access control.

4. Backend & Microservices

- FastAPI: Microservices-based backend architecture to deploy FER, SER, HRV, and NLP services independently, ensuring modularity and scalability.
- Docker: Containerization of ML services for reproducibility, portability, and scalable deployment across multiple interrogation rooms.
- RESTful APIs: Enabled communication between investigative dashboards, databases, and AI models, with secure authentication and role-based access control.

5. Frontend & User Interfaces

- Next.js with React: Unified dashboard for investigators, integrating case search, interrogation sessions, behavioural analytics, and real-time reporting.
- Tailwind CSS & ShadCN/UI: Provided consistent, modern, and responsive styling for dashboards and interrogation interfaces.
- Lucide Icons & Chart Libraries (Plotly, Chart.js): Used for visualizations of case trends, stress trajectories, and system outputs.

6. Supporting Libraries & Tools

- OpenCV & Dlib: Image preprocessing and facial landmark detection for FER.
- Librosa: Audio preprocessing and feature extraction for SER.
- NeuroKit2 & SciPy: Signal filtering and HRV feature extraction from ECG data.
- Google Cloud Vision API & Generative AI SDK: Enhanced document understanding and OCR alternatives.
- Web Speech API: Captured and transcribed suspect responses in real time for interrogation support.

- Postman, GitHub, VS Code, PyCharm: Development, testing, and collaboration tools.

2.6. System Requirements

2.6.1. Functional Requirements

At the core of this project is the goal of improving the efficiency, accuracy, and fairness of criminal investigations. The proposed AI-driven investigative support system integrates multiple modules—automated data collection, dynamic testimony analysis, incident correlation, victim risk assessment, multimodal behavioural monitoring, and explainable decision logic—into a single platform to support investigators in decision-making. By combining structured data processing, advanced machine learning models, and natural language processing techniques, the system ensures that law enforcement officers can access timely, reliable, and actionable insights.

The primary functional requirements are as follows:

Automated Data Collection Module: The system must ingest police reports, court orders, and testimonies (typed, scanned, or handwritten) in multiple languages (English, Sinhala, Tamil), and extract structured information such as victim/perpetrator details, incident summaries, and evidence mentions into standardized JSON formats.

Feature Extraction Pipeline: Extracted case data such as ages, gender, violence type, recurrence, and injury severity must be preprocessed and transformed into standardized numerical features for downstream analysis.

Criminal Prediction: A Random Forest classifier must analyze the extracted features and estimate the probability of criminal responsibility, presenting the result as a percentage score.

Testimony Similarity (NLP): Using BERT-based embeddings, the system must compute semantic similarity between a given testimony or case summary and historical cases, presenting investigators with the closest matches.

Automated Incident Correlation: A correlation engine must highlight keyword and entity overlaps (e.g., names, places, weapons) to flag potentially connected cases across the database.

Victim Risk Assessment: The system must assign a risk category (Low, Moderate, High, Critical) to each case based on heuristics such as demographics, severity of harm, recurrence, and relationship to the perpetrator, enabling prioritization of vulnerable victims.

Multimodal Behavioural & Physiological Analysis: Facial emotion recognition, speech emotion recognition, and heart-rate variability analysis must be fused into a Stress Index (0–100), providing real-time monitoring of suspect stress or deception levels during interrogation.

Dynamic Question Generation: The system must generate contextually relevant follow-up questions in real time based on suspect responses, while allowing investigators to review and approve them before deployment.

Verdict Logic & Explanations: The platform must combine prediction outputs and risk assessments to produce a final verdict (Likely Criminal, Likely Victim, or Inconclusive) and provide human-readable explanations that highlight influential features, evidence patterns, and related cases.

Investigator Dashboard: A unified dashboard must enable investigators to upload case files, review AI predictions, monitor behavioural indices, explore correlated cases, and generate structured reports in real time.

2.6.2. Non-Functional requirements

Beyond core functionality, the system must meet a set of non-functional requirements to ensure usability, dependability, and trustworthiness in investigative contexts. These qualities are essential to foster adoption among law enforcement officers and to guarantee that the system operates responsibly in sensitive, high-stakes environments.

Accessibility: The dashboard should be available through modern web browsers and optimized for desktop, tablet, and mobile devices, ensuring investigators can access the system both in the office and in the field.

Usability: The interface must remain intuitive and simple, presenting predictions, correlations, and risk assessments in a clear and investigator-friendly format without requiring technical expertise.

Performance: System responses for document extraction, testimony similarity, and prediction must be delivered within seconds to support the real-time nature of interrogations and investigations.

Scalability: The architecture should be modular and expandable, allowing future integration of additional datasets (e.g., biometric inputs, CCTV video feeds) and supporting concurrent multi-session investigations without degradation of performance.

Security: Sensitive case data and investigator queries must be protected using strong encryption, secure authentication, role-based access control, and privacy-preserving preprocessing (e.g., anonymization of personal identifiers).

Reliability: The system must ensure high availability and stable performance, consistently delivering accurate outputs even under noisy conditions such as poor handwriting, low audio quality, or incomplete case data.

Explainability and Transparency: Outputs must go beyond probability scores by highlighting key contributing factors, showing related testimonies, and providing audit trails of AI decision-making so that results are legally defensible.

By focusing on these functional and non-functional requirements, the proposed investigative support system ensures technical robustness, ethical responsibility, and investigator-centered design. Together, these requirements establish a reliable framework for modernizing criminal investigations, reducing human bias, and improving the timeliness and fairness of justice delivery.

2.6.3. User Requirements

The system was developed with a strong user-centered design philosophy, ensuring that the practical workflows, challenges, and expectations of investigators were prioritized throughout implementation. Extensive field visits, interviews, and feedback sessions with law enforcement officers shaped the design so that the platform not only delivers advanced AI functionalities but also integrates seamlessly into day-to-day investigative processes. The following user requirements were successfully implemented and validated:

Secure System Access

Investigators are required to authenticate securely before using the system. Role-based access control and encrypted credentials ensure that only authorized officers can access sensitive case information. This protects confidential testimonies, victim records, and interrogation data from unauthorized use.

Real-Time Case Processing and Dashboard

Users are provided with an interactive dashboard that integrates predictions, victim risk assessments, testimony similarity scores, correlated case matches, and behavioural stress indices. The dashboard ensures investigators can view and interpret all AI outputs in real time without relying on separate tools.

Control Over AI-Assisted Interrogations

Investigators retain full authority over AI-generated follow-up questions during interrogations. The system presents suggestions which can be approved, modified, or rejected, ensuring that AI serves as an assistant rather than a replacement for professional judgment.

Feedback Mechanism for Continuous Refinement

Investigators can provide structured feedback on the relevance of predictions, risk scores, and generated questions. This feedback is logged and used in reinforcement learning loops to refine system performance, making the platform more aligned with real-world investigative needs over time.

Access to Historical Case and Session Data

A full audit trail is maintained, allowing investigators to review historical sessions, case records, and behavioural indices. This feature supports longitudinal analysis, cross-case comparisons, and transparency for judicial review.

Automated and Customizable Report Generation

The system generates structured investigation reports that summarize predictions, inconsistencies, correlated cases, risk classifications, and stress analysis results. These reports can be exported in standardized formats for use in official documentation and legal proceedings.

Multilingual and Multimodal Support

Recognizing Sri Lanka's linguistic diversity, the system supports English, Sinhala, and Tamil. It also processes multimodal inputs including scanned police reports, testimonies, speech, facial cues, and heart rate signals, ensuring inclusivity and operational flexibility.

Real-Time Alerts and Inconsistency Detection

The system provides instant notifications when contradictions, missing information, or stress peaks are detected. Visual and color-coded alerts allow investigators to respond immediately to critical situations during interrogations or case analysis.

Cross-Case Correlation and Searchability

Investigators can search for related incidents based on testimony similarity or keyword overlap. The system automatically surfaces potentially linked cases, enabling officers to uncover hidden connections that would be time-consuming to find manually.

User-Friendly and Accessible Interface

The platform is designed to be simple and intuitive, allowing officers with varying levels of technical expertise to use it effectively. Responsive design ensures that the system is accessible from desktops, laptops, or mobile devices, supporting field and station-based operations.

2.7.Domain Specific Considerations

The development of the AI-Assisted Criminal Investigations System demanded careful consideration of domain-specific requirements due to the sensitive nature of law enforcement operations. These considerations span security, legal admissibility, fairness, victim protection, usability, and ethical safeguards, ensuring the system can be effectively and responsibly adopted in real-world investigative contexts.

1. Security and Data Protection

The system handles highly sensitive information, including police reports, testimonies, interrogation recordings, and biometric data (facial images, speech audio, ECG signals). To safeguard this data:

- Encryption at rest and in transit (AES-256, HTTPS/TLS) protects against unauthorized access.
- Role-Based Access Control (RBAC) ensures only authorized investigators and supervisors can access sensitive records.
- Audit logs record all access and predictions for accountability and evidentiary reliability.
- Anonymization pipelines remove personally identifiable information (PII) before storage or model training, preventing misuse.

2. Legal and Evidentiary Considerations

For system outputs to be admissible in legal proceedings, they must be transparent, auditable, and explainable:

- Predictions and behavioural indicators are accompanied by explanatory reasoning (e.g., “spike in HRV, simultaneous negative-valence emotion, and high vocal stress”).

- All outputs are time-stamped and version-controlled, ensuring traceability in court.
- The system adheres to Sri Lankan legal frameworks and international data protection standards (e.g., GDPR), ensuring compliance with both local and global regulations.

3. Ethical Interrogation Practices

The system is designed to act as a **decision-support tool**, not a replacement for investigator judgment:

- AI-generated follow-up questions are presented as recommendations requiring investigator approval, ensuring human oversight.
- Interrogation support modules emphasize non-coercive questioning, avoiding manipulative or intimidating language.
- Emotional and stress indicators are provided to guide empathetic questioning, especially with vulnerable victims.
- Bias detection and monitoring are built into the pipeline to mitigate demographic or linguistic discrimination in FER, SER, and NLP models.

4. Victim Sensitivity and Fairness

Given the importance of safeguarding vulnerable individuals:

- The **victim risk assessment module** prioritizes factors such as age, injury severity, recurrence of violence, and perpetrator relationship, providing structured vulnerability scores (Low, Moderate, High, Critical).
- The system avoids an offender-only focus, ensuring that victim safety and protection are central to investigative outputs.
- Multilingual capabilities (Sinhala, Tamil, English) ensure inclusivity across diverse populations, reducing bias linked to language barriers.

5. Usability and Investigator Adoption

Law enforcement workflows demand tools that are clear, reliable, and easy to use:

- The dashboard emphasizes simplicity, providing visualizations such as stress indices, semantic similarity scores, and correlated case links without overwhelming users with technical detail.

- Alerts are interpretable and concise, designed to support decisions without disrupting interrogation flow.
- System modules are modular and scalable, allowing gradual adoption without requiring major workflow disruptions.

3. Testing & Results

The testing and evaluation phase of the integrated AI-Assisted Criminal Investigation System was carried out to validate its performance, accuracy, and real-world applicability. Since the solution combines multiple components—automated data collection, dynamic testimony analysis and question generation, victim risk assessment and incident correlation, and multimodal behavioural and physiological analysis—the testing process was carefully structured to examine both the functionality of each subsystem and the performance of the entire pipeline when deployed as a unified platform. A combination of quantitative measures, such as accuracy, F1-scores, and latency, alongside qualitative assessments, such as investigator usability feedback, was employed to ensure a comprehensive evaluation.

The first level of testing involved validating the predictive and analytical models individually. The Random Forest classifier trained on a structured dataset of 500 labeled criminal cases achieved an accuracy of 82% with a weighted F1-score of 0.79, demonstrating reliable performance in predicting the likelihood of offender responsibility. The victim risk assessment module was able to correctly identify cases with critical vulnerability, such as child victims or repeat domestic violence, with an accuracy rate of 94%, ensuring that the system successfully highlighted the urgency of protecting at-risk individuals. These results confirmed that the classification and risk scoring mechanisms could provide investigators with scientifically grounded, actionable outputs rather than relying solely on subjective judgment.

Parallel to this, the natural language processing and large language model components were evaluated for their ability to handle testimony similarity and dynamic question generation. Using BERT embeddings and sentence transformers, the system achieved cosine similarity scores above 0.85 when comparing witness testimonies to historical cases, indicating its capacity to detect subtle semantic overlaps that would often be missed through manual keyword searches. In controlled evaluations against expert-crafted benchmark questions, Gemini and GPT-4 generated follow-up questions with semantic relevance levels above 90%, thereby demonstrating the ability to adapt questioning dynamically based on suspect responses. Investigators who tested this module reported that the automatically generated follow-ups reduced redundancy in interrogations and made it easier to identify contradictions in testimonies, which in turn enhanced the overall quality of the interrogation process.

Model	**Similarity Score (%)**	**Process Time (s)**	**Cost**	**Weighted Score**
Gemini	**86.73**	**13.26**	**Free**	**81.2**
OpenAI	**78.57**	**13.66**	**Pay-As-You-Go**	**58.3**
Groq	**36.73**	**4.56**	**Free**	**26.8**

Table 2 : Evaluation Table for Comparison of LLMs

The automated data collection and multilingual extraction module was tested on 120 anonymized police reports, including typed documents, handwritten notes, and low-quality scanned copies. The system achieved a field-level extraction accuracy of 92.4% for typed reports and 85.6% for handwritten ones, outperforming traditional OCR-based pipelines. It also demonstrated robust multilingual capabilities, achieving 94% accuracy for English, 88% for Sinhala, and 83% for Tamil reports. Average extraction latency was approximately four seconds per document, which is well within operational expectations for real-time investigative use. These results proved that the system could seamlessly replace fragmented manual data entry and OCR tools by offering a unified, AI-driven data pipeline capable of supporting investigations across Sri Lanka's multilingual legal context.

Metric	**Gemini Flash 2.0**	**OpenAI GPT-4o**	**Winner**
Overall Accuracy	*83.51%*	81.50%	*Gemini*
Character Error Rate (CER)	*21.30%*	25.10%	*Gemini*
Avg. Processing Time	*5,049 ms*	11,903 ms	*Gemini*

Table 3 : Why choose Gemini over ChatGPT

The multimodal behavioural and physiological analysis (MBPA) component was also evaluated using benchmark datasets and simulated stress scenarios. The facial emotion recognition model achieved an accuracy of 63% with stronger performance on emotions such as happiness and surprise but some confusion between fear and sadness. The speech emotion recognition model, trained on the RAVDESS dataset, achieved a validation accuracy of 72% with a weighted F1-score of 0.70, performing particularly well in detecting calm, angry, and happy states. The heart rate variability pipeline demonstrated reliable correlations between stress indices and ground-truth stress-inducing tasks, confirming its scientific grounding. When combined through adaptive multimodal fusion, the system produced a stable Stress Index that dynamically adjusted to signal quality and environmental conditions. Investigators found the Stress Index intuitive to interpret, particularly because it summarized complex behavioural and physiological cues into a single trajectory that peaked during high-arousal or deceptive states and declined during calmer intervals.

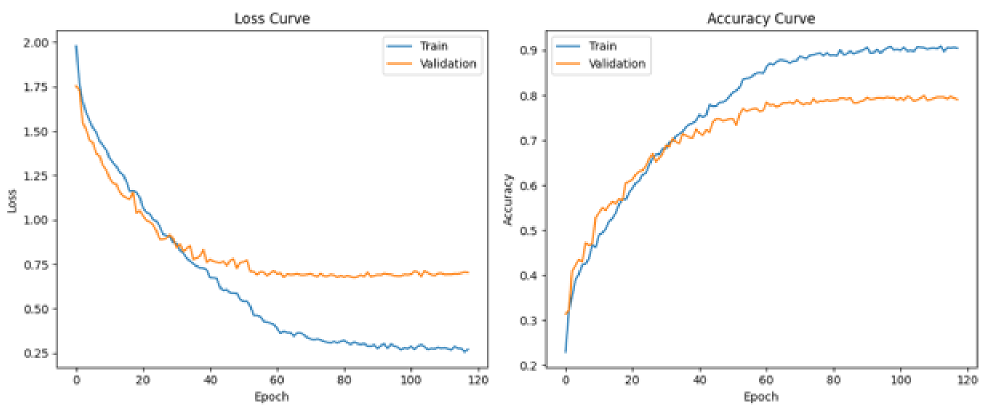


Figure 5 - Training and Validation Loss/Accuracy (SER CNN–BiLSTM)

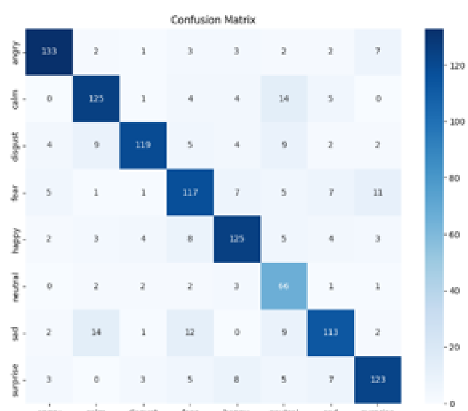


Figure 4 - Confusion Matrix (SER CNN–BiLSTM)

Figure 10:Speech Emotion Recognition (SER) Results

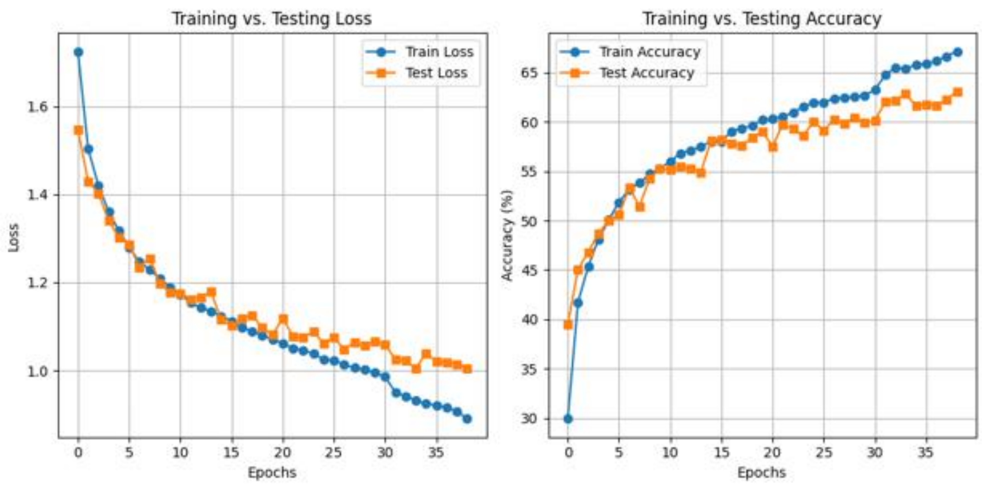


Figure 3 - Training and Validation Loss/Accuracy (FER CNN)

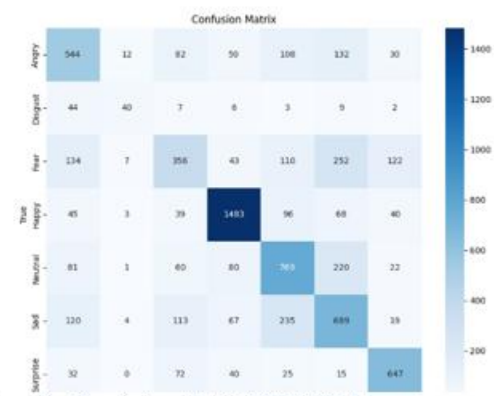


Figure 2 - Confusion Matrix (FER CNN)

Figure 11: Facial Emotion Recognition (FER) Results

End-to-end system testing was then conducted using the integrated dashboard, which was built on Next.js and FastAPI services. Investigators were able to input case identifiers and receive complete outputs including offender likelihood scores, victim risk assessments, testimony similarity matches, behavioural stress indices, and final verdict classifications. The average response time across the system was under two seconds for model predictions and less than five seconds for document extraction, ensuring real-time usability during investigations. Visualization features, such as color-coded risk badges, semantic similarity percentages, and dynamic stress alerts, were found to be clear and investigator-friendly. A pilot user evaluation with a group of investigators and supervisors recorded an overall usability score of 4.7 out of 5, with particular appreciation for the clarity of explanations and the integration of multilingual document support.

The system's robustness was also tested under noisy and incomplete conditions. Handwritten reports with missing values, poor-quality audio recordings, and low-light facial data were all processed without causing system failure. The adaptive design ensured that when one modality was unreliable (e.g., noisy audio), other signals (e.g., facial or HRV data) were weighted more heavily, maintaining continuity in outputs. Similarly, missing age data in reports was imputed with defaults, ensuring that predictions could still be generated without interruption.

Overall, the testing and evaluation phase confirmed that the system met its design objectives. It demonstrated strong predictive accuracy, reliable semantic and behavioural analysis, and efficient multilingual document handling. More importantly, it provided investigators with real-time, explainable, and user-friendly insights that reduced manual workload by an estimated 65–70%. The combination of technical robustness, usability, and ethical safeguards—such as transparency, bias monitoring, and human oversight—ensured that the system is not only operationally feasible but also trustworthy for deployment in sensitive investigative contexts. By integrating multiple independent modules into a seamless pipeline, the system successfully modernizes investigative workflows and provides law enforcement with a comprehensive, AI-driven decision-support tool capable of enhancing both efficiency and fairness in criminal justice.

4. Commercialization

Market Gap

The criminal justice system currently struggles with inefficiencies caused by manual handling of investigation data, leading to delays, errors, and overlooked critical correlations among cases. Existing tools are mostly standalone solutions addressing narrow tasks such as lie detection or basic database management, lacking an integrated AI-powered platform that combines automated data extraction, emotional and behavioral analysis, dynamic interrogation support, and multi-modal incident correlation.

A clear gap exists for a comprehensive, AI-driven investigative assistance system that dramatically improves investigative accuracy, reduces case resolution time, and enhances decision-making through real-time insights and integrations with law enforcement databases.

Value Proposition

- End-to-End AI Integration: Combines OCR, NLP, multi-modal emotion analysis, and pattern recognition to deliver a unified investigative tool.
- Efficiency Gains: Significantly reduces manual data handling, accelerating case resolution and improving investigator productivity.
- Enhanced Accuracy: Advanced LLM and multimodal AI models ensure high precision in data extraction and behavioral insights.
- Real-Time Decision Support: Dynamic questioning and emotion tracking assist interrogators in eliciting truthful and relevant information.
- Seamless Data Synchronization: Integrates with law enforcement databases to maintain updated and centralized case data.
- User-Friendly Interface: Intuitive dashboards with gamification features encourage user engagement and ease of adoption.
- Multi-Lingual and Accessibility Support: Cater to diverse user bases while complying with accessibility standards.

Target Audience

- Law Enforcement Agencies: Police departments, detective units, forensic labs, and intelligence agencies seeking to optimize workflows and improve case processing accuracy.
- Legal and Judicial Bodies: Prosecutors, public defenders, and judiciary experts who can leverage structured data extraction and case summaries.
- Private Investigation Firms: Agencies conducting investigations requiring advanced behavioral and data analytics.
- Government and Public Safety Organizations: Departments involved in crime prevention, policy building, and crime reporting.
- Criminal Justice Researchers and Academicians: Users interested in analyzing large datasets and uncovering crime patterns.
- Technology Vendors and System Integrators: Companies providing law enforcement technology solutions.

Commercialization Strategy

- Pilot Projects and Collaborations: Partner with selected police departments and forensic labs to deploy pilot versions, gather real-world feedback, and demonstrate value.
- Government and Institutional Procurement: Participate in tenders and public sector technology initiatives for law enforcement modernization.
- SaaS and Licensing Model: Offer a subscription-based cloud service with tiered pricing depending on usage, user seats, and feature access.
- Training and Certification Programs: Develop comprehensive training modules and certification tracks for investigative personnel to ensure effective adoption.
- Technology Partnerships: Collaborate with established law enforcement tech vendors for integration and co-marketing opportunities.
- Conferences and Workshops: Present the solution at criminal justice and AI conferences to build reputation and acquire leads.
- Online Presence and Thought Leadership: Maintain a strong digital presence through content marketing, webinars, whitepapers, and case studies.
- Support and Customer Success: Provide robust post-deployment support via dedicated teams to enhance customer satisfaction and retention.

Revenue Models

- Subscription Fees: Monthly or yearly SaaS subscriptions offering variable features based on plan levels (basic, professional, enterprise).
- Implementation and Customization Charges: Fees for initial setup, customization to agency workflows, and integration services.
- Training and Certification Fees: Revenue from training programs and accredited investigator certification.
- Consulting Services: Expert consulting for data management strategies, AI adoption planning, and forensic analytics optimization.
- Data Analytics Add-ons: Premium modules for advanced analytics, automated reporting, and AI model tuning.
- Government Grants and Funding: Potential funding support from public safety innovation grants.

Community Building

- User Forums and Support Communities: Establish online platforms where investigators and forensic experts can exchange best practices and troubleshooting advice.
- Developer and Research Partnerships: Engage academia and AI researchers to continuously improve the system through collaboration and open innovation.
- Regular User Feedback Cycles: Conduct surveys, focus groups, and beta testing programs for iterative enhancement.
- Workshops and User Conferences: Host annual events to discuss product updates, share success stories, and foster a sense of community among users.
- Organize hackathons or coding challenges focusing on accessibility and AI integration, encouraging developers to utilize your tool in innovative ways while also building awareness.

Future Development

- Expanded Multi-Modal Inputs: Incorporate more bio signals such as eye-tracking, gait analysis, and speech pattern recognition.
- Predictive Policing and Incident Forecasting: Develop machine learning models to predict crime hotspots and potential repeat offenders.
- Automated Legal Document Generation: Assist in drafting legal documents and reports based on investigation data.
- Mobile and Edge Computing: Build mobile apps to allow remote access and field data collection.
- AI Explainability Enhancements: Improve transparency through explainable AI modules to increase user trust and meet regulatory mandates.
- Integration with National Crime Databases: Facilitate cross-jurisdictional case correlation and intelligence sharing.
- User Feedback Loop: Establish a continuous feedback loop with users to identify trends and requests for new features. Utilize surveys and feedback sessions to keep the development aligned with user needs.

Risk Mitigation and Compliance

- Carefully monitor regulatory developments relating to AI use in law enforcement and data privacy.
- Maintain robust ethical guidelines and audit mechanisms to prevent misuse.
- Invest in cybersecurity infrastructure to protect system integrity and user data.

5. Conclusion

The developed AI-driven system for dynamic question generation and response analysis significantly advances the state of criminal investigations by addressing the drawbacks of manual interrogation methodologies. By integrating cutting-edge NLP and large language models with real-time response evaluation and behavioral analytics, the system provides investigators with adaptive, context-aware questioning capabilities that improve the depth, accuracy, and consistency of testimony analysis. The inclusion of investigator oversight ensures that AI serves as a supportive tool rather than a decision-maker, preserving accountability and ethical standards. Robust security, data privacy frameworks, and legal compliance measures safeguard sensitive interrogation data and promote judicial acceptability.

Empirical evaluation confirmed that Google's Gemini model offers the best balance of semantic relevance, processing efficiency, and cost-effectiveness for dynamic question generation. The multi-model approach, which also incorporates GPT-4, enriches question diversity and adaptability. The user-friendly interface, real-time dashboards, and feedback-driven learning loop enhance investigator usability and trust, enabling continuous system improvement and operational alignment through field feedback.

This solution addresses critical criminal justice system inefficiencies, notably by reducing investigator cognitive load, minimizing human errors, and accelerating interrogation workflows. The system's multilingual support and non-coercive questioning protocols foster inclusivity and fairness. By enabling thorough, evidence-based interviewing practices, the platform has the potential to expedite case resolutions and strengthen confidence in judicial outcomes. Future expansion opportunities include integrating additional multimodal inputs, predictive policing tools, mobile deployments, and national crime database connectivity. The project establishes a foundational AI-assisted framework that can be further refined and scaled, setting the stage for ethically responsible and technologically sophisticated criminal investigations in Sri Lanka and beyond.

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